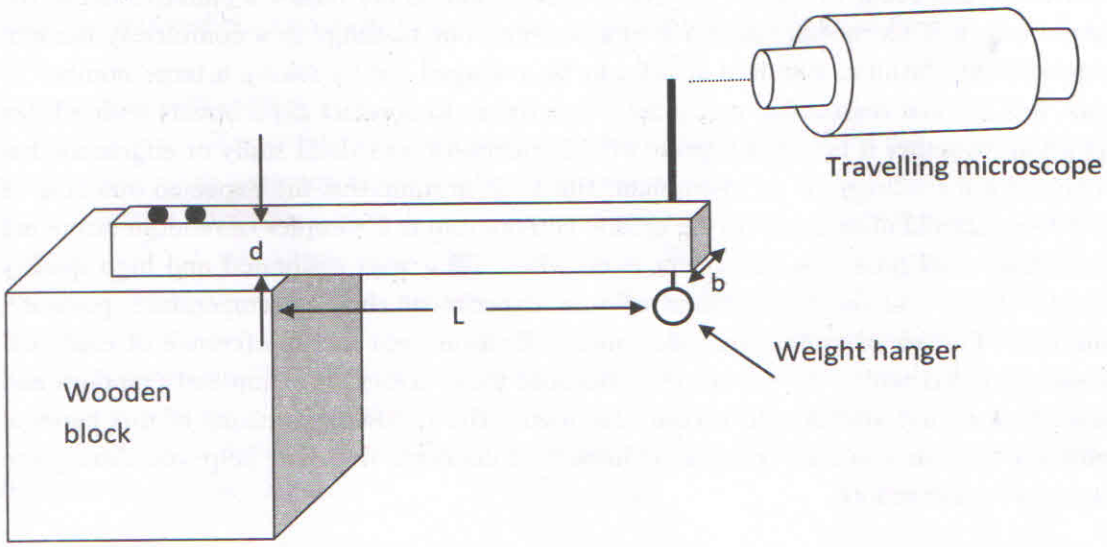
SINGLE CANTILEVER

OBSERVATIONS:

Experimental Setup:



Formula:

The Young's modulus of the material of the cantilever is calculated using the formula;

$$q = \frac{4mgL^3}{bd^3\delta_{mean}} \quad Nm^{-2}$$

Where, m = mass for which depression is measured

L is the length of the cantilever in metre

b is the breadth of the cantilever in metre

d is the thickness of the cantilevering metre

δ is the mean depression for =a mass m (40g) in metre

g is the acceleration due to gravity = 9.8 m/s

Least count of the travelling microscope

$$L.C = \frac{\text{Value of 1 main scale division}}{\text{Total number of vernier scale divisions}} = \frac{0.5}{50} = 0.001 \text{ c m}$$

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Table 1: Travelling microscope readings for the load increasing.

Load (g)	MSR (cm)	CVD	TR= MSR + (CVD x LC) (cm)
W	5.80	13	5.813
W + 10	5.70	31	5.731
W + 20	5.60	46	5.646
W + 30	5.55	01	5.551
W + 40	5.45	21	5.471
W + 50	5.35	30	5.380
W + 60	5.25	46	5.296
W + 70	5.20	02	5.202

Table 2: Travelling microscope readings for the load decreasing.

Load (g)	MSR (cm)	CVD	TR= MSR + (CVD x LC) (cm)
W + 70	5.20	02	5.202
W + 60	5.30	03	5.303
W + 50	5.35	29	5.379
W + 40	5.45	17	5.467
W + 30	5.50	49	5.549
W + 20	5.60	40	5.640
W + 10	5.70	29	5.729
W	5.80	10	5.810

Table 3: Depression for a load of 40g (difference column)

Load (g)	TM reading (cm)		Mean TMR $R_L = (R_1 + R_2)/2$	Load (g)	TM readings (cm)		Mean TMR $R_R = (R_3 + R_4)/2$	Depression for 40g ($\delta = R_L \sim R_R$) cm
	Load increasing R_1	Load Decreasing R_2			Load Increasing R_3	Load Decreasing R_4		
W	5.810	5.813	5.8115	W + 40	5.471	5.467	5.469	0.3425
W + 10	5.731	5.729	5.730	W + 50	5.380	5.379	5.3795	0.351
W + 20	5.646	5.640	5.643	W + 60	5.296	5.303	5.2995	0.34
W + 30	5.551	5.549	5.550	W + 70	5.202	5.202	5.202	0.348

Mean depression for a load of 40 g = δ = 0.3453 cm.

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Length of the cantilever, L = 17.8 cm.

Table 4: Breadth of the cantilever using vernier callipers

$$L.C = \frac{\text{Value of 1 main scale division}}{\text{Total number of vernier scale divisions}} = \frac{0.1}{10} = 0.01 \text{ cm}$$

Trial No.	MSR (cm)	CVD	TR = MSR + (CVD x LC) (cm)
1	2.3	5	2.35
2	2.3	6	2.36
3	2.3	4	2.34
4	2.3	5	2.35

Mean Breadth b = 2.35 cm

Table 5: Thickness (d) of the cantilever using screw gauge.

Least Count of Screw Gauge:

$$\text{Pitch} = \frac{\text{Distance moved on the pitch scale}}{\text{Total number of rotations given to screwhead}} = \frac{5}{5} = 1 \text{ mm}$$

$$L.C = \frac{\text{Pitch of the screw gauge}}{\text{Total number of head scale divisions}} = \frac{1}{100} = 0.01 \text{ mm}$$

Zero error (ZE) = −4

Trial No.	PSR (mm)	HSD	TR= PSR + (HSD – (ZE)) LC in mm
1	0	79	0.83
2	0	82	0.86
3	0	84	0.88
4	0	83	0.87

Mean thickness, d = 0.85 mm

ERROR CALCULATION:

The Young's modulus of the material of the cantilever is calculated using the formula;

$$q = \frac{4mgL^3}{bd^3\delta_{mean}} \quad Nm^{-2}$$

Taking the logarithm, $\log q = \log(4mg) + 3\log L - \log b - 3\log d - \log \delta$

Taking Partial Derivatives and Using Modulus:

The relative error (or fractional error) in q is given by:

$$\frac{\Delta q}{q} = 3\frac{\Delta L}{L} + \frac{\Delta b}{b} + 3\frac{\Delta d}{d} + \frac{\Delta \delta}{\delta}$$

Substitute the least count values of length (L), breadth (b), thickness (d) and depression (δ) as ΔL, Δb, Δd, *and* $\Delta \delta$ respectively to calculate the value of $\Delta q/q$.

CALCULATIONS:

Table 1: 5.80 + 13×0.001 = 5.813 cm
5.70 + 31×0.001 = 5.731 cm
5.60 + 46×0.001 = 5.646 cm
5.55 + 01×0.001 = 5.551 cm

Table 5.45 + 21×0.001 = 5.471 cm
5.35 + 30×0.001 = 5.380 cm
5.25 + 46×0.001 = 5.296 cm
5.20 + 02×0.001 = 5.202 cm

Table 2: 5.20+02×0.001 = 5.202 cm
5.30+03×0.001 = 5.303 cm
5.35+29×0.001 = 5.379 cm
5.45+01×0.001 = 5.467 cm

5.50+49×0.001 = 5.549 cm
5.60+40×0.001 = 5.640 cm
5.70+29×0.001 = 5.729 cm
5.80+10×0.001 = 5.810 cm

Table 3 = $\delta_{avg} = \frac{0.3425+0.351+0.34+0.348}{4} = 0.3453$ cm

Table 4

2.3 + 5(0.01) = 2.35 cm	2.3 + 4(0.01) = 2.34 cm
2.3 + 6(0.01) = 2.36 cm	2.3 + 5(0.01) = 2.35 cm

$q = \frac{4mgL^3}{bd^3\delta_{mean}} = \frac{4\times40\times9.8\times(17.8\times10^{-2})^3}{2.35\times10^{-2}\times(0.85\times10^{-3})(0.3453)\times10^{-2}}$

$= 1.9059\times10^{11}\ Nm^{-2}$

$\frac{\Delta q}{q} = \frac{3(0.1)}{17.8} + \frac{0.01}{2.35} + \frac{0.01}{0.85} + \frac{0.001}{0.3453}$

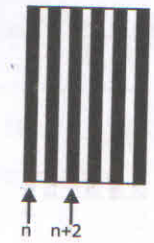
$= 0.0592$

Result:

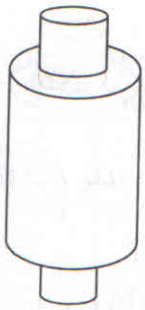
1. Young's modulus of the given material is found to be 1.9059×10^{11} N m^{-2}.
2. The relative error (or fractional error) in q is 0.0592.

INTERFERENCE AT A WEDGE

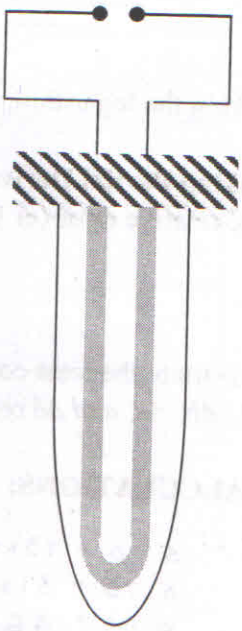
OBSERVATIONS



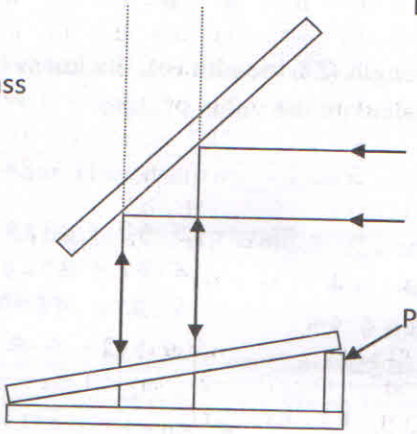
Fringe pattern system



Travelling microscope



Sodium vapor lamp



Inclined glass plate

Least count of the travelling microscope

$$LC = \frac{\text{Value of 1 main scale division}}{\text{Total number of vernier scale divisions}} = \frac{0.05}{50} \text{ cm}$$

Tabular column

Fringe number	TM readings, MSR + (CVD x LC)	Total reading (cm)
N	4.4 + (12 x 0.001)	4.412
n + 2	4.5 + (05 x 0.001)	4.505
n + 4	4.6 + (25 x 0.001)	4.625
n + 6	4.7 + (04 x 0.001)	4.704
n + 8	4.8 + (44 x 0.001)	4.844
n + 10	4.95 + (35 x 0.001)	4.985
n + 12	5.1 + (18 x 0.001)	5.118
n + 14	5.25 + (04 x 0.001)	5.254
n + 16	5.35 + (46 x 0.001)	5.396
n + 18	5.5 + (31 x 0.001)	5.531
n + 20	5.65 + (19 x 0.001)	5.669
n + 22	5.8 + (24 x 0.001)	5.824

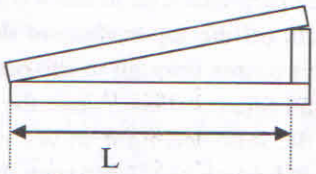
Table 2:

Fringe No.	TM Reading (cm), R_1	Fringe No.	TM Reading (cm), R_2	$X = 12\beta = R_2 - R_1$ (cm)
N	4.412	n + 12	5.118	0.706
n + 2	4.505	n + 14	5.254	0.749
n + 4	4.625	n + 16	5.396	0.771
n + 6	4.704	n + 18	5.531	0.827
n + 8	4.844	n + 20	5.669	0.825
n + 10	4.985	n + 22	5.824	0.839

Average x = 0.786×10^{-2} m

The fringe width, β, is x/12 = 0.0655×10^{-2} m.

To measure the length, L, of the air wedge



L = 0.069 m

The thickness of the paper is calculated by using the formula

$t = \frac{\lambda L}{2\beta}$ = 31.878×10^{-2} m

Where,

t is the thickness of the paper 31.278×10^{-6} m

L is the length of the air wedge 0.069 m

λ is the wavelength of sodium light = 589.3nm

β is the fringe width 0.0655×10^{-2} m

Calculations

$$t = \frac{\lambda L}{2\beta} = \frac{589.3 \times 10^{-9} \times 0.069}{2 \times 10^{-2} (0.0655)} = 31.278 \times 10^{-6} \text{ m}$$

Result:

The thickness of the given paper = 31.278×10^{-6} m.

If white light is used instead of monochromatic light, the fringes appear coloured because different wavelengths interfere at different positions.

The fringe width (distance between consecutive fringes) 'β' is given by: $\beta = \lambda / 2\theta$

where: λ is the wavelength of the light, θ is the angle of the wedge (the angle between the two glass plates).

As θ is small it can be expressed as: $\theta = t / L$

where: 't' is the thickness of the paper or spacer, and 'L' is the length of the air wedge.

Thus, the fringe spacing β can be rewritten as: $\beta = \lambda L / 2t$

Therefore, the thickness of the paper (spacer) can be determined using: $t = \lambda L / 2\beta$

Procedure:

a) Experimental Setup

- The given optically plane glass plates and the reflecting glass plate are cleaned well by using muslin cloth before starting the experiment. The optically plane glass plates are kept one above the other. A thin rectangular paper piece is inserted between the glass plates at one extreme end, such that the piece of paper is perpendicular to the length of the glass plate. Now a wedge-shaped air film is formed in between the plates. The thickness of the air film gradually increases, from the point of contact of glass plate, till the inner edge of the paper piece.
- Now illuminate the air wedge by a parallel beam of sodium light using reflecting glass plate by making an angle of inclination 45° such that light after reflection will fall on the optical flats as shown in the diagram. The whole setup is kept below the objective of the travelling microscope such that the illuminated portion of the air wedge comes under the field of view of travelling microscope. Then focus the travelling microscope using rack and pinion screw such that the alternate dark and bright interference fringes are clearly seen.

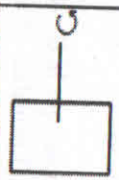
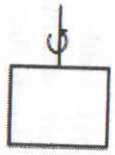
b) Measurement of fringe width

- The least count of travelling microscope is determined by finding the ratio of the value of one main scale division to the total number of vernier scale divisions.
- The vertical cross wire is adjusted to be at the center of any one of the dark fringes. The reading on the main scale and the coinciding vernier scale division are noted. Let this fringe be the nth dark fringe.
- The reading corresponding to the dark fringes n+2, n+4, n+6 up to n+22 are noted. The microscope may be moved either only to the left or the right side for this purpose.
- The readings are tabulated in the observation column and the fringe width is calculated.
- The length of the air wedge between the point of contact of the glass plates and the inner edge of the paper piece is measured using traveling microscope / scale.

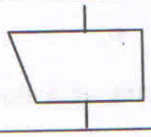
- The thickness of the given paper is calculated using the formula. $t = \frac{\lambda L}{2\beta}$ m

Result: The thickness of the given paper = 31.278×10^{-6} m.

TORSION PENDULUM – OBSERVATIONS

Body	M (Kg)	Dimension (m)	Axis	Moment of Inertia I (Kgm²)	Time for 10 osc. (s)	Mean time 't' for 10 oscillation. (s)	Period T=t/10 (s)	$\frac{I}{T^2}$ (Kgm²s⁻²)
Circular plate	1.468	Radius = 0.0765		$I_1 = \frac{MR^2}{2}$ = 4.2956×10^{-3}	1. 43.85 2. 44.22 3. 44.24	44.103	T_1 = 4.4103	$\frac{I_1}{T_1^2}$ = 2.2084×10^{-4}
				$I_2 = \frac{MR^2}{4}$ = 2.1478×10^{-3}	1. 31.50 2. 31.63 3. 31.20	31.4433	T_2 = 3.144	$\frac{I_2}{T_2^2}$ = 2.1723×10^{-4}
Rectangular plate	1.290	Length, L = 0.153		$I_3 = \frac{M(L^2+B^2)}{12}$ = 2.7341×10^{-3}	1. 41.18 2. 40.98 3. 41.09	41.0833	T_3 = 4.1083	$\frac{I_3}{T_3^2}$ = 1.619×10^{-4}
				$I_4 = \frac{ML^2}{12}$ = 2.5165×10^{-3}	1. 33.84 2. 34.09 3. 33.97	33.9660	T_4 = 3.3966	$\frac{I_4}{T_4^2}$ = 2.1813×10^{-4}
		Breadth, B = 0.045		$I_5 = \frac{MB^2}{12}$ = 2.2176×10	1. 23.22 2. 23.25 3. 23.09	23.1867	T_5 = 2.3187	$\frac{I_5}{T_5^2}$ = 0.4048×10^{-4}

$\left(\frac{I}{T^2}\right)$ =

Axis through CG	Time for 10 sec. (s)	Mean time(t) for 10 oscillations in s	Period T=t/10 s
	1. 37.41 2. 37.40 3. 37.28	3.7363	T_α = 3,7363
	1. 20.60 2. 20.38 3. 20.63	20.5367	T_β = 2.05367

Determination of Moment of Inertia of Irregular body

Formulae:

(a) Moment of inertia of irregular body about the axis through the CG and perpendicular to its plane, I'

$$I_\alpha = \left(\frac{I}{T^2}\right)_{mean} \times T_\alpha^2 \quad Kgm^2$$

(b) Moment of inertia of the irregular body about the axis through the CG and parallel to its plane

$$I_\beta = \left(\frac{I}{T^2}\right)_{mean} \times T_\beta^2 \quad Kgm^2$$

Determination of rigidity modulus of the material of the wire

Radius of the wire (r) using screw gauge.

$$\text{Pitch} = \frac{\text{Distance moved on the pitch scale}}{\text{Total number of rotations given to screwhead}} = \frac{5}{5} = 1 \text{ mm}$$

$$\text{L.C} = \frac{\text{Pitch of the screw gauge}}{\text{Total number of head scale divisions}} = \frac{1}{100} = 0.01 \text{ mm}$$

Zero error (ZE) = -18.

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Paste the data sheets here

Trial No.	PSR(mm)	HSD	TR= PSR + (HSD – (ZE))LC (mm)
1	0	54	0.72
2	0	55	0.73
3	0	56	0.74
4	0	56	0.74

Mean diameter of the wire, d = 0.7325 mm 0.7325×10^{-3} m

Mean Radius of the wire, r = 0.3663×10^{-3} m

Length of the wire, L = 0.298 m

Rigidity modulus $\eta = \left(\frac{8\pi L}{r^4}\right)\left(\frac{I}{T^2}\right)$

$= 8.5087 \times 10^{10}$ N/m^2

Result:

1. The moment of inertia of irregular body about an axis perpendicular to the plane $I_\alpha = 2.855 \times 10^{-3}$ Kgm^2.

2. The moment of inertia of irregular body about an axis parallel to the plane

   $I_\beta = 8.5087 \times 10^{-4}$ Kgm^2.

3. Rigidity modulus of the material of the wire 'η' = 8.5087×10^{10} N/m^2

CALCULATIONS:

$I_1 = \frac{mR^2}{2} = \frac{1.468 \times (0.0765)^2}{2} = 4.2959 \times 10^{-3}$ Kgm^3 $\frac{I_1}{T_1^2} = 2.2084 \times 10^{-4}$ Kgm^{2}s^{-2}

$I_2 = \frac{mR^2}{2} = \frac{1.468 \times (0.0765)^2}{2} = 2.1478 \times 10^{-3}$ Kgm^3 $\frac{I_2}{T_2^2} = 2.1723 \times 10^{-4}$ Kgm^{2}s^{-2}

$T_3 = \frac{M(L^2+B^2)}{12} = \frac{1.290 \times (0.153^2 + 0.045^2)}{12} = 2.7341 \times 10^{-3}$ Kgm^2 $\frac{I_3}{T_3^2} = 1.6190 \times 10^{-4}$ Kgm^{2}s^{-2}

$I_4 = \frac{ML^2}{12} = \frac{1.290 \times (0.153)^2}{12} = 2.5165 \times 10^{-3}$ Kgm^3 $\frac{I_4}{T_4^2} = 2.1813 \times 10^{-4}$ Kgm^{2}s^{-2}

$I_5 = \frac{MB^2}{12} = \frac{1.290 (0.045)^2}{12} = 0.2176 \times 10^{-3}$ Kgm^3 $\frac{I_4}{T_4^2} = 0.4048 \times 10^{-4}$ Kgm^{2}s^{-2}

$I_\alpha = \left[\frac{I}{T^2}\right]_{mean} \times T_\alpha^2 = 2.0453 \times 10^{-4} \times 3.37363 = 2.8551 \times 10^{-3}$ Kgm^3

1. TR = 0 + (54 - (-18)) × 0.01 mm = 0.72 mm = 0.72×10^{-3} m
2. TR = 0 + (55 - (-18)) × 0.01 mm = 0.73 mm = 0.73×10^{-3} m
3. TR = 0 + (56 - (-18)) × 0.01 mm = 0.74 mm = 0.74×10^{-3} m
4. TR = 0 + (56 - (-18)) × 0.01 mm = 0.74 mm = 0.74×10^{-3} m

$\eta = \left[\frac{8\pi L}{r^4}\right]\left[\frac{I}{T^2}\right]_{mean} = \frac{8 \times 3.14 \times 0.298}{(0.3663 \times 10^{-3})^4} \times 2.0453 \times 10^{-4}$

$= 8.5087 \times 10^{10}$ N/m^2

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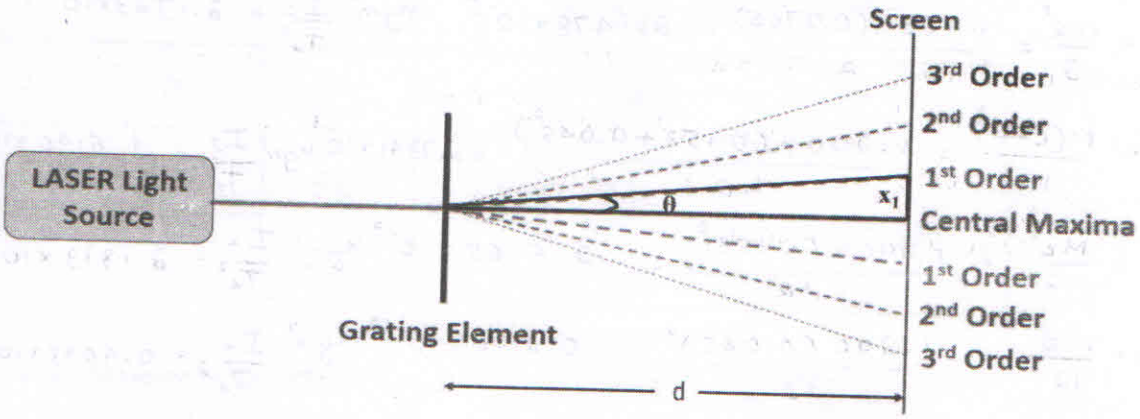
Result:

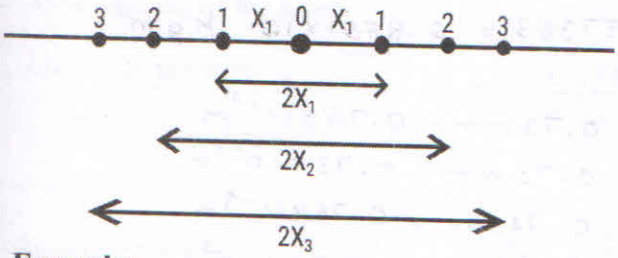
1. The moment of inertia of irregular body about an axis perpendicular to the plane $I_\alpha = 2.855 \times 10^{-3}$ Kgm^2.
2. The moment of inertia of irregular body about an axis parallel to the plane $I_\beta = 8.629 \times 10^{-4}$ Kgm^2.
1. Rigidity modulus of the material of the wire 'η' = 8.5087×10^{10} N/m^2

LASER DIFFRACTION

OBSERVATIONS:

Diagram:





Diffraction angle θ_n, $\tan\theta_n = \frac{x_n}{d}$

X_n = Distance between the central maximum and the nth order secondary maximum.

d = Distance between grating and the screen

Formula:

Wavelength of Laser source $\lambda = \frac{C \sin\theta_n}{n}$ ..... m

Where C is the grating constant, n is the order of the maximum, θ is the angle of diffraction

Grating Constant: $C = \frac{1\,\text{inch}}{\text{No. of lines (N) per inch}} = \frac{2.54\times10^{-2}\,m}{500} = 5.08\times10^{-5}\,\text{m}$

Distance between the grating and the screen, d = 121.8 cm

Table:

Diffraction order (n)	Distance $2X_n$(cm)	Distance X_n(cm)	Diffraction angle (θ_n in degree) $\theta_n = \tan^{-1}\left(\frac{X_n}{d}\right)$	Wavelength λ (nm) $\lambda = \frac{C \sin\theta_n}{n}$	Fractional error $\left(\frac{\delta\lambda}{\lambda}\right)$
1.	3.3	1.65	0.9761	688.0903	0.0606
2.	5.2	3.1	1.4579	646.2372	0.0322
3.	10.1	5.05	2.3742	701.4760	0.0198
4.	13.3	6.65	3.1251	692.3561	0.0150
5.	16.8	8.4	3.9516	700.6387	0.0119
Average wavelength of the laser and fractional error				$\lambda =$ 685.7597 nm	$\left(\frac{\delta\lambda}{\lambda}\right) = 0.0279$

Error Formula:

The formula used wavelength of laser light is $\lambda = \frac{C \sin\theta_n}{n}$

Taking the logarithm, $\log \lambda = \log (C/n) + \log(\sin\theta)$

Simplifying, we get the relative uncertainty (or fractional uncertainty) in λ is given by:

$$\frac{\delta\lambda}{\lambda} = \frac{\delta x_n}{x_n} + \frac{x_n \delta x_n + d\delta d}{\left(x_n^2 + d^2\right)} \approx \frac{\delta x_n}{x_n}$$

Substitute the least count values of (x_n) distance between the central maxima and the order of diffraction and (d) distance between grating and the screen as δx_n and δd, respectively to calculate the value of $\delta\lambda / \lambda$.

Calculations:

1. $\theta_1 = \tan^{-1}\left(\frac{1.65}{121.8}\right) = 0.9761$ $\quad \lambda_1 = \frac{C \sin\theta_1}{1} = 688.0903$
2. $\theta_2 = \tan^{-1}\left(\frac{3.1}{121.8}\right) = 1.4579$ $\quad \lambda_2 = \frac{C \sin\theta_2}{2} = 646.2372$
3. $\theta_3 = \tan^{-1}\left(\frac{5.05}{121.8}\right) = 2.3742$ $\quad \lambda_3 = \frac{C \sin\theta_3}{3} = 701.4760$
4. $\theta_4 = \tan^{-1}\left(\frac{6.65}{121.8}\right) = 3.1251$ $\quad \lambda_4 = \frac{C \sin\theta_4}{4} = 692.3561$
5. $\theta_5 = \tan^{-1}\left(\frac{8.4}{121.8}\right) = 3.9516$ $\quad \lambda_5 = \frac{C \sin\theta_5}{5} = 700.6387$

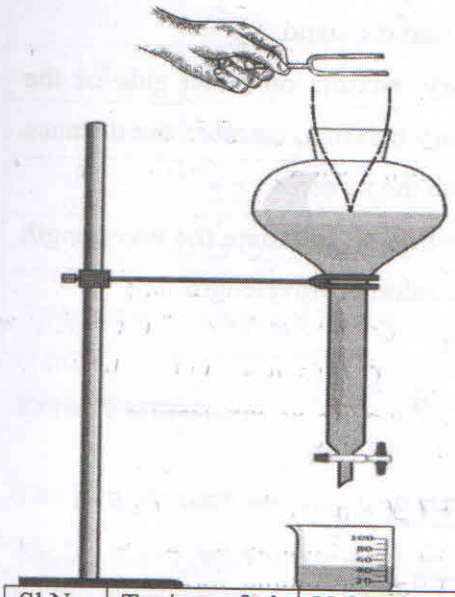
Result: 1. The wavelength of laser light is found to be 685.7597 nm

2. The relative error (or fractional error) in λ is 0.0279

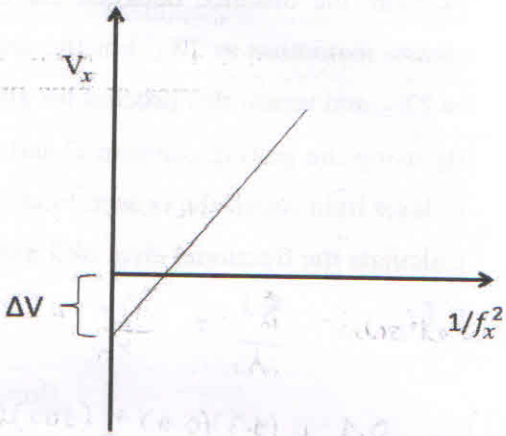
VOLUME RESONATOR

OBSERVATIONS

Experimental setup:



Model Graph



Sl.No.	Tuning fork frequency f in Hz	Volume of resonating air (water collected) in m^3			1/ f^2	$f^2.(V+\Delta V)$	$[f^2 (V+\Delta V)]_{avg}$
		V_1 in cm^3	V_2 in ~~m^3~~ cm^3	Mean volume V in ~~m^3~~ cm^3	(10^{-6})	in ~~$m^3 s^{-2}$~~ $cm^3 s^{-2}$	
1	480	154	150	152	4.3403	32256000	
2	426	192	194	193	5.5104	32847156	31482300
3	384	212	210	211	6.7817	29343744	
4	$f_{unknown}$	136	134	V_f 134	-	$(V_f+\Delta V)$ = 122	

Determining unknown frequency of the tuning fork

$$[f^2 (V+\Delta V)]_{avg} = f^2_{unknown} (V_f+\Delta V)_{unknown}$$

$$f_{unknown} = \sqrt{\frac{[f^2 (V+\Delta V)_{avg}]}{V_f + \Delta V}} \quad Hz = 507.9878 \text{ Hz}$$

Where, $f_{unknown}$ is the unknown frequency of the tuning fork, f = Known frequency of the tuning fork, V = volume of air resonating with the fork of known frequency, V_f = volume of air resonating with the fork of unknown frequency and ΔV is the volume correction obtained from graph.

Calculations:

$f^2 (V+\Delta V)$

$f_1^2 (V+\Delta V)_1 = 480^2 (152 - 12) = 32256000 \text{ cm}^3 \text{s}^{-2}$

$f_2^2 (V+\Delta V)_2 = 426^2 (193 - 12) = 32847156 \text{ cm}^3 \text{s}^{-2}$

$f_3^2 (V+\Delta V)_3 = 384^2 (211 - 12) = 29343744 \text{ cm}^3 \text{s}^{-2}$

$$f_{unknown} = \sqrt{\frac{[f^2(V+\Delta V)]_{avg}}{(V_f + \Delta V)}} \text{ Hz}$$

$$= \sqrt{\frac{31482300}{122}}$$

$$= 507.9878 \text{ Hz}$$

Need for volume correction:

When the Helmholtz resonator is excited, the air in the cavity is compressed and rarefied as the air in the neck moves back and forth. The volume of air inside the cavity that actually contributes to the oscillatory motion is not limited to the geometric volume of the cavity. Some of the air just outside the cavity near the neck also gets involved in the oscillation due to the pressure fluctuations. This means that the effective volume of the resonator is slightly larger than the simple geometric volume.

Procedure: The aspirator bottle is filled with water. The tuning fork of known frequency is excited and it is held above the mouth of the aspirator bottle as shown in the figure. At the same time, water in the bottle is allowed to run out of aspirator bottle by opening the pinch cock attached to the out let tube. Water is allowed to flow out till the air in the resonator is in resonance with the tuning fork used. In the correct position of resonance, a note of maximum intensity will be heard. At the same moment the flow of water should be stopped using the pinch cock. The volume 'V' of water is collected in a beaker and measured using measuring jar. The volume is equal to the volume of air in the resonator, which is in resonance with the tuning fork, when vibrated. The experiment is repeated for the same tuning fork and the mean value of volume 'V' is found. The experiment is repeated for different tuning fork as above and volume is noted. The volume of air 'V_f' is found out with the fork of unknown frequency $f_{unknown}$.

Plot a graph of volume 'V' along y-axis and $1/f^2$ along x-axis and find out the value of y-intercept 'ΔV' from the resulting graph.

Formula:

$$f_{unknown} = \sqrt{\frac{[f^2\,(V + \Delta V)_{avg}]}{V_f + \Delta V}} \quad Hz$$

Where, $f_{unknown}$ is the unknown frequency of the tuning fork, f = Known frequency of the tuning fork, V = volume of air resonating with the fork of known frequency, V_f = volume of air resonating with the fork of unknown frequency and ΔV is the volume correction obtained from graph.

Result:

Volume correction = ΔV = -12

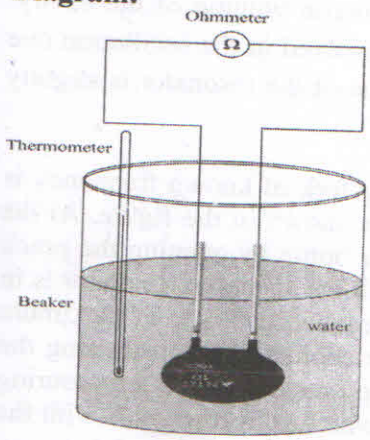
The unknown frequency of given tuning fork = 507.9878 Hz.

Note: Start the experiment by choosing higher frequency tuning fork and proceed the experiment in decreasing order of their frequencies.

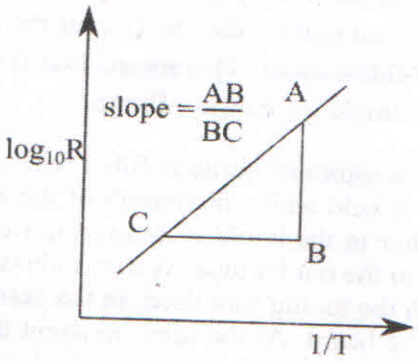
BAND GAP OF A THERMISTOR

OBSERVATIONS:

Diagram:



Model Graph:



Formula: $E_g = \dfrac{4.606 \times k \times m}{1.6 \times 10^{-19}}$ eV

Where E_g = Energy gap of a given thermistor in eV

k = Boltzmann constant = 1.381×10^{-23} J/K

m = Slope of the graph

Table:

Sl. No	Temp t °C	Temp T(K)	R (Ω)	log R	1/T ($\times 10^{-3}$ K^{-1})
Room Temp	27.0	300.15	53.1	1.7251	3.3316
1.	81	354.15	8.4	0.9243	2.8216
2.	80	353.15	8.6	0.9345	2.8316
3.	79	352.15	8.8	0.9445	2.8396
4.	78	351.15	9.1	0.9590	2.8478
5.	77	350.15	9.3	0.9685	2.8559
6.	76	349.15	9.6	0.9820	2.8641
7.	75	348.15	9.8	0.9912	2.8723
8.	74	347.15	10.1	1.0043	2.8806
9.	73	346.15	10.5	1.0211	2.8889
10.	72	345.15	10.8	1.0334	2.8972

CALCULATIONS:

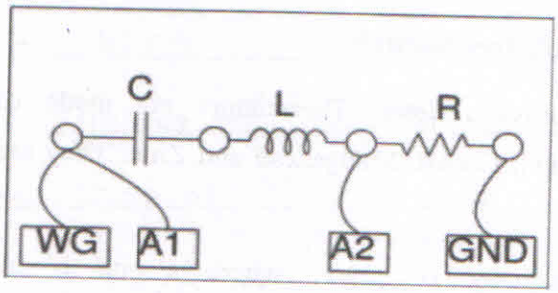
$$E_g = \frac{4.606 \times k \times m}{1.6 \times 10^{-19}} = \frac{4.606 \times 1.381 \times 10^{-23} \times 1.425 \times 10^{-3}}{1.6 \times 10^{-19}} = 0.5665 \text{ eV}$$

Result: The energy gap (band gap) of the given thermistor is 0.5665 eV.

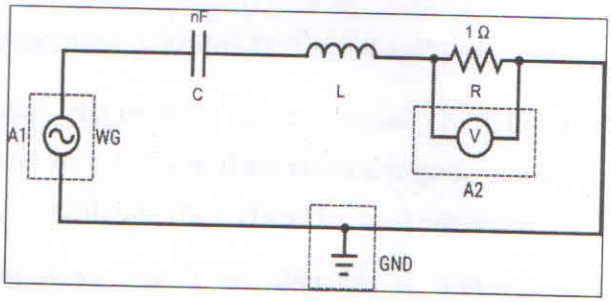
SERIES L-C-R CIRCUIT

OBSERVATIONS:

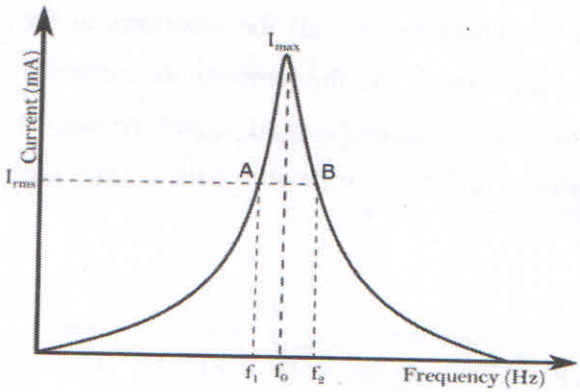
ExpEyes circuit:



Conventional Circuit:



Model Graph:



Formulae

Capacitance C = 474 nF

$L = \frac{1}{4\pi^2 f_0^2 C}$ = 1.1035 H

I_{max} = 21.94 mA

$I_{RMS} = \frac{I_{max}}{\sqrt{2}}$ = 15.51 mA

Resonant Frequency f_0 = 220.06 Hz

Lower Cut off frequency f_1 = 204.165 Hz

Upper Cut off frequency f_2 = 235.575 Hz

Band width $\Delta f = f_2 - f_1$ = 31.41 Hz

Quality Factor $Q = \frac{f_0}{\Delta f}$ = 7.00617

Calculations:

$L = \frac{1}{4\pi^2 (220.064)^2 \times 474 \times 10^{-9}} = 1.1035$ H

Band width = $f_2 - f_1$ = 235.575 - 204.165 = 31.41 Hz

Quality factor = $\frac{f_0}{\Delta f} = \frac{220.064}{31.41} = 7.00617$

Note: Graph plot procedure using GNUPLOT

Go to respective folder/directory on the desktop.

Right click on the mouse, **select open in terminal**

type `gnuplot` **in the terminal and press ↵ Enter**

Sample code

```
set timestamp
set title 'Series LCR circuit-Rajesh'
set xlabel 'Frequency in Hz'
set ylabel 'Current in mA'
plot 'lcr.txt' w lp
```

Marking points on the graph

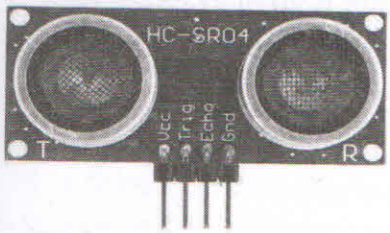
- Take the cursor to the tip of the curve
- Press the scroll button on the mouse
- Note down the corresponding frequency (x co-ordinate) as resonant frequency f_0 and the corresponding current(y co-ordinate) as maximum or resonant current I_{max}
- Find out I_{RMS} $(=\frac{I_{max}}{\sqrt{2}})$
- Press the scroll button on the mouse at I_{RMS} on the left side of the graph and note down the corresponding frequency as lower cut off frequency f_1.
- Press the scroll button on the mouse at I_{RMS} on the right side of the graph and note down the corresponding frequency as upper cut off frequency f_2.
- Export the file as PDF

Result:

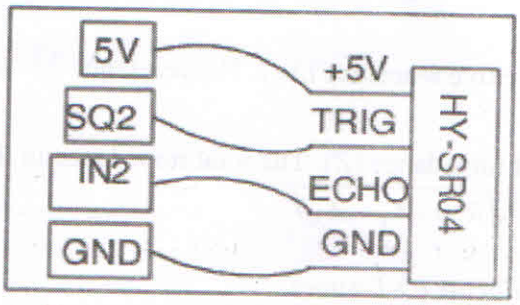
1.	Resonant frequency of the circuit f_0	220.64 Hz
2.	Self-inductance of the given coil L	1.1034 H
3.	Quality factor from graph Q	7.0062
4.	Band width $\Delta f = f_2 - f_1$	31.41 Hz

SPRING CONSTANT

OBSERVATIONS

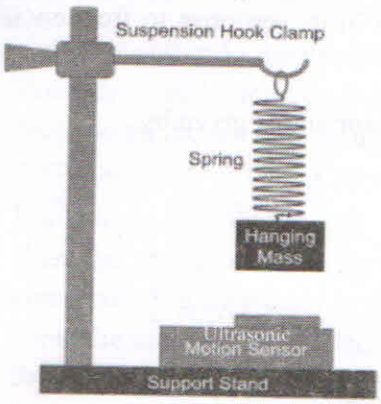


Ultrasonic sensor (HC-SR04)

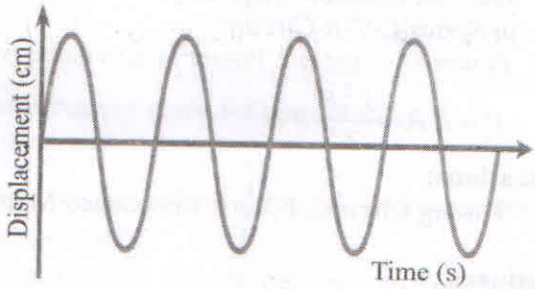


Connections with Exp-eyes device

Experimental setup



Model graph



Formula: Spring constant is given by $K = 4\pi^2 \frac{m}{T^2}$ N/m

Where, *m* is the mass of the load (kg). T is the time period (s) and K is the spring constant N/m.

Graph plotting with GNUPLOT

Go to respective folder in which the file is saved.

Right click at the empty area, select **'open in terminal'** and Run the following commands:

```
gnuplot
set timestamp
set title 'Spring Constant-Raj'
set xlabel 'Time in S'
set ylabel 'Displacement in cm'
set multiplot layout 2,2 columns
plot 'spring1.txt' w lp
plot 'spring2.txt' w lp
plot 'springseries.txt' w lp
plot 'springparallel.txt' w lp
```

SPRING CONSTANT

OBSERVATIONS

Table 1: Spring constant K_1 for the given material

(Mass of the hanger + Mass of the slotted weights) = m_1 = _____ Kg

Trial No	Frequency(Hz)	Time for one osc T_1 (s)	T_1^2	Spring constant $K_1 = 4\pi^2 \frac{m_1}{T_1^2}$ (N/m)
1	1.37	0.73	0.5329	7.4082

Table2: Spring constant K_2 for the given material

(Mass of the hanger + Mass of the slotted weights) = m_2 = _____ Kg

Trial No	Frequency(Hz)	Time for one osc T_2 (s)	T_2^2	Spring constant $K_2 = 4\pi^2 \frac{m_2}{T_2^2}$ (N/m)
1	1.37	0.73	0.5329	7.4082

Table 3: Spring constant K_S in series combination for the given material

(Mass of the hanger + Mass of the slotted weights) = m_s = _____ Kg

Trial No	Frequency(Hz)	Time for one osc T_S (s)	T_S^2	Spring constant $K_S = 4\pi^2 \frac{m_s}{T_s^2}$ (N/m)	$K_{eff} = \frac{K_1 K_2}{K_1 + K_2}$ (N/m)	Error = $K_{eff} \sim K_S$ (N/m)
1	1.33	0.75	0.5625	3.5091	3.7041	0.1950

Table 4: Spring constant K_P in parallel combination for the given material

(Mass of the Scale + Mass of the hanger + Mass of the slotted weights) = m_P = _____ Kg

Trial No	Frequency(Hz)	Time for one osc T_P (s)	T_P^2	Spring constant $K_P = 4\pi^2 \frac{m_p}{T_P^2}$ (N/m)	$K_p = K_1 + K_2$ (N/m)	Error = $K_{eff} \sim K_P$ (N/m)
1	1.47	0.68	0.4628	14.6678	14.8164	0.1486

Results:

a)	The spring constant for the given springs are	K_1 = 7.4082 N/m K_2 = 7.4082 N/m
b)	The spring constant in series combination is	Ks = 3.7041 N/m
c)	The spring constant in parallel combination is	Kp = 14.8164 N/m

About hardware and software:

The circuit consists of a microcontroller system, an ultrasonic sensor, Expeyes hardware and a spring mass system. A microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system. An ultrasonic sensor measures the distance of an object using sound waves. The ultrasonic sensor consists of 4 pins (Voltage pin, Trigger Pin, ECHO pin, GND pin). Connections are made as shown in the figure, voltage pin is connected to +5V of the expEYES-17; the trig pin of the sensor is connected to the SQ2 of the expEYES-17; the echo pin is connected to the IN2; GND pin is connected to the GND of the expeyes-17. The expeyes-17 interfaced with CPU of the computer. It combines the real-time measurement capability of micro-controllers with the ease and flexibility of Python programming language for data analysis and visualization.

CALCULATIONS:

$T_1^2 = (0.73)(0.73) = 0.5329\ s^2$

$T_2^2 = (0.73)(0.73) = 0.5329\ s^2$

$T_3^2 = (0.75)(0.75) = 0.5265\ s^2$

$T_4^2 = (0.68)(0.68) = 0.4624\ s^2$

$K_1 = \frac{4\pi^2(0.1)}{0.5329} = 7.4082\ N/m$

$K_2 = \frac{4\pi^2(0.1)}{0.5329} = 7.4082\ N/m$

$K_3 = \frac{4\pi^2(0.05)}{0.5625} = 3.5091\ N/m$

$K_p = \frac{4\pi^2(0.1718)}{0.4624} = 14.8664\ N/m$

$K_s = \frac{K_1 K_2}{K_1 + K_2} = \frac{(7.4082)(7.4082)}{7.4082+7.4082} = 3.7041\ N/m$, error = $0.1950\ N/m$

$K_p = K_1 + K_2 = 7.4082 + 7.4082 = 14.8164$ error = $0.1486\ N/m$

10

R.V. COLLEGE OF ENGINEERING®

OBSERVATION / DATA SHEET

Date 5-12-2024 Name Pramod Varamballi

Dept./Lab Physics Class _____ Expt./No. 7

Title Single cantilever

Aim: To determine the Young's of the material of the single cantilever.

Apparatus: Thick rectangular metallic strip, slotted weight with hanger, travelling microscope, screw gauge. Vernier callipers. and metre scale.

RVCE DEPARTMENT
EVALUATION OF EXPT...

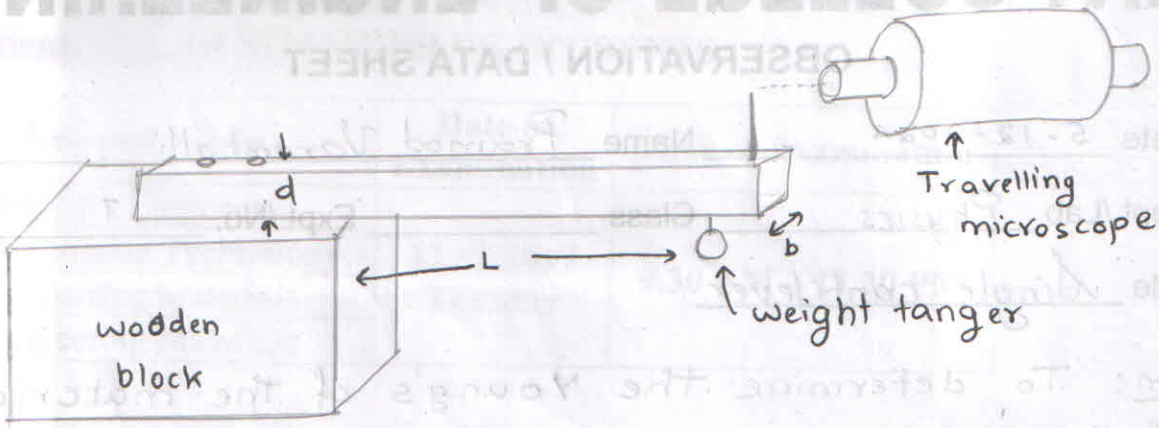
Scheme of Evaluation	Course Outcome	Max Marks	Marks Scored	Initial of ... Staff
Data Sheet and Experimental Set up (4+6)	CO1	10	4+6	
Conduction of Experiment	CO2	10	10	
Substitution, Calculation & Accuracy	CO3	10	10	
Total Marks Scored		30	30	
Signature of the Faculty			30	

26/12/24

Signature of
Teacher Incharge

04.12.2017

Observations :-



Formula :

The Young's modulus of materials of the canti is calculated using the formula.

$$q = \frac{4mgL^3}{bd^3\delta_{mean}} \quad Nm^{-2}$$

m = mass of which depression
L = length of the cantilever (m)
b = breadth of the cantilever (m)
d = Thickness of the cantilever (m)
δ = mean depression for a mass m (40g) in cm
g = is acceleration due to gravity = 9.8 m/s

$$L.C = \frac{\text{Value of 1 main scale division}}{\text{Total no of vernier scale division}} = \frac{0.5}{50} = 0.001 \text{ cm}$$

R.V. C

Date

Dept./Lab

Title

load (g)	MSR
W	5.
W+10	5.
W+20	5.
W+30	5.
W+40	5.
W+50	5.
W+60	5.
W+70	5.

Table 2 : Tra

load (g)	MSR
W+70	5.2
W+60	5.
W+50	5.
W+40	5.
W+30	5.
W+20	5.
W+10	5.
W	5.8

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OBSERVATION / DATA SHEET

Date _____ Name _____

Dept./Lab _____ Class _____ Expt./No. _____

Title _____

Table 1: Travelling microscope reading for increasing load

load (g)	MSR (cm)	CVD	TR = MSR + (CVD × LC) cm.
W	5.80	13	5.813
W+10	5.70	31	5.731
W+20	5.60	46	5.646
W+30	5.55	01	5.551
W+40	5.45	21	5.471
W+50	5.35	30	5.380
W+60	5.25	46	5.296
W+70	5.20	02	5.202

Table 2: Travelling microscope reading for decreasing load

load (g)	MSR (cm)	CVD	TR = (MSR) + (CVD × LC)
W+70	5.20	02	5.202
W+60	5.30	03	5.303
W+50	5.35	29	5.379
W+40	5.45	17	5.467
W+30	5.50	49	5.549
W+20	5.60	40	5.640
W+10	5.70	29	5.729
W	5.80	10	5.810

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Table 3 : Depression for a load of 40 g

Load (g)	TM reading (cm) load increasing R_1	TM reading (cm) load decreasing R_2	Mean TMR $R = \frac{(R_1+R_2)}{2}$	load (g)	TM readings (cm) load increasing R_3	TM readings (cm) load decreasing R_4	Mean TMR $R = \frac{(R_3+R_4)}{2}$	Dep for 4 $\delta = $
W	5.810	5.813	5.8115	W+40	5.471	5.467	5.469	0.3
W+10	5.731	5.729	5.730	W+50	5.380	5.379	5.3795	0.3
W+20	5.646	5.640	5.643	W+60	5.296	5.303	5.2995	0.3
W+30	5.551	5.549	5.550	W+70	5.202	5.202	5.202	0.3

mean depression for load of 40g = δ = 0.3453

length of cantilever, L = 17.8 cm

Table 4 : Breadth of Cantilever using vernier callipers

$LC = \frac{\text{Values of 1 main scale division}}{\text{Total no of vernier scale division}} = \frac{0.1}{10} = 0.01$

Trial No	MSR	CVD	TR = MSR + (CVD x LC) cm
1	2.3	5	2.35
2	2.3	6	2.36
3	2.3	4	2.34
4	2.3	5	2.35

mean Breadth b = 2.35 cm

Table 5 : Thickness (d) of the cantilever using screw gauge.

$Pitch = \frac{\text{Distance moved on the pitch scale}}{\text{Total number of rotations given to screwhead}}$

$LC = \frac{\text{Pitch of screw}}{\text{Total number of head scale division}} = \frac{1}{100} =$ 0.01 cm

Zero error = (ZE) = −4

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Dept./Lab

Title

Trial No	PSR C
1	0
2	0
3	0
4	0

ror = $\frac{\Delta q}{q}$ = 3

alculations

ple 1 =

= 5.80 + 1.3

= 5.70 + 31

= 5.60 + 4

= 5.55 + 01

le R

= 5.20 + 02 x

= 5.30 + 03

= 5.35 + 29

= 5.45 + 17 x

rrage = 0.3425

= 0.3

	Mean TMR	Dep for δ
R_4	$R_R = (R_3 + R_4)$	$\delta = 1$
7	5.469	0.3
9	5.3995	0.3
3	5.2995	0.3
2	5.202	0.3

δ = 0.3453

callipers

1 / 0 = 0.01

LC) cm

cm

using

screwhead

= 0.045 cm

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OBSERVATION / DATA SHEET

Date _____ Name _____

Dept./Lab _____ Class _____ Expt./No. _____

Title _____

Trial No	PSR (mm)	HSD	TR = PSR + (HSD - ZE).LC in mm
1	0	79	0.83
2	0	82	0.86
3	0	84	0.88
4	0	83	0.87

mean thickness, d = 0.85 mm

ror = $\frac{\Delta q}{q} = 3\frac{\Delta l}{l} + \frac{\Delta b}{b} + 3\frac{\Delta d}{d} + \frac{\Delta \delta}{\delta}$

alculations

ple 1 =

= 5.80 + 1.3 × 0.001 cm = 5.813 cm

= 5.70 + 31 × 0.001 = 5.731 cm

= 5.60 + 46 × 0.001 = 5.646 cm

= 5.55 + 01 × 0.001 = 5.551 cm

= 5.45 + 21 × 0.001 = 5.471 cm

5.35 + 30 × 0.001 = 5.380 cm

5.25 + 46 × 0.001 = 5.296 cm

5.20 + 02 × 0.001 = 5.202 cm

le R

= 5.20 + 02 × 0.001 = 5.202 cm

= 5.30 + 03 × 0.001 = 5.303 cm

= 5.35 + 29 × 0.001 = 5.379 cm

= 5.45 + 17 × 0.001 = 5.467 cm

= 5.50 + 49 × 0.01 = 5.549

= 5.60 + 40 × 0.01 = 5.640

= 5.70 + 29 × 0.01 = 5.729

= 5.80 + 10 × 0.01 = 5.810

e 3

verage = $\frac{0.3425 + 0.351 + 0.34 + 0.348}{4}$

= 0.3453

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Table 4

1. $TR_1 = 2.3 + 5(0.01) = 2.35\,cm$ $TR_3 = 2.3 + 4(0.01) = 2.34$

$TR_2 = 2.3 + 6(0.01) = 2.36\,cm$ $TR_4 = 2.3 + 5(0.01) = 2.35$

mean breadth = 2.35cm

Table 5

$TR = 0 + (79+4)0.01 = 0.83\,mm$ $TR_3 = 0 + (84+4)0.01 = 0.88\,mm$

$TR_2 = 0 + (82+4)0.01 = 0.86\,mm$ $TR_4 = 0 + (83+4)0.01 = 0.87\,mm$

mean thickness = 0.85 mm

$$q = \frac{4mgL^3}{bd^3\delta_{mean}} = \frac{4 \times 40 \times 9.8 \times (17.8)^3 \times 10^{-6}}{2.35 \times 10^{-2} \times (0.85 \times 10^{-3})^3 (0.3453) \times 10^{-2}}$$

$$\Rightarrow \frac{4 \times 40 \times 9.8 \times (17.8)^3}{2.35 \times (0.83)^3 \times 0.3453} \times \frac{10^{-6}}{10^{-2} \times 10^{-9}}$$

$$\Rightarrow \frac{8843131.136}{0.4639794201} \times 10^2$$

$$\Rightarrow 1905931.7.62 \times 10^2 = 19.0593 \times 10^{11}$$

$$= 1.9059 \times 10^{11}$$

$$\frac{\Delta q}{q} = 3 \times \frac{0.1}{17.8} + \frac{0.01}{2.35} + 3 \times \frac{0.01}{0.85} + \frac{0.001}{0.3453}$$

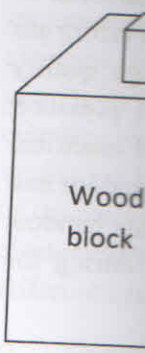
$$= 0.0168 + \text{p.}$$

$$\frac{\Delta q}{q} = 0.0592$$

Result:

1) Young's modulus of given material is 1.9059×10^{11} N

2) The relative error in q is 0.0592.



Formula:

The Young's

Where, m =

L is th

b is th

d is th

δ is th

g is th

Least count o

L.C =

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R.V. COLLEGE OF ENGINEERING®

OBSERVATION / DATA SHEET

Date 26-12-2024 Name Pramod Varamballi

Dept./Lab Physics Class 1 Expt./No. 8

Title Interference at wedge

Aim: To determine the thickness of a thin paper piece by setting up interference fringes at an air wedge.

Apparatus :- Two . Thick glass plates, sodium vapour lamp. travelling microscope. a thin paper piece, reflecting glass plate reading lens and a scale.

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EVALUATION OF EXPERIMENT

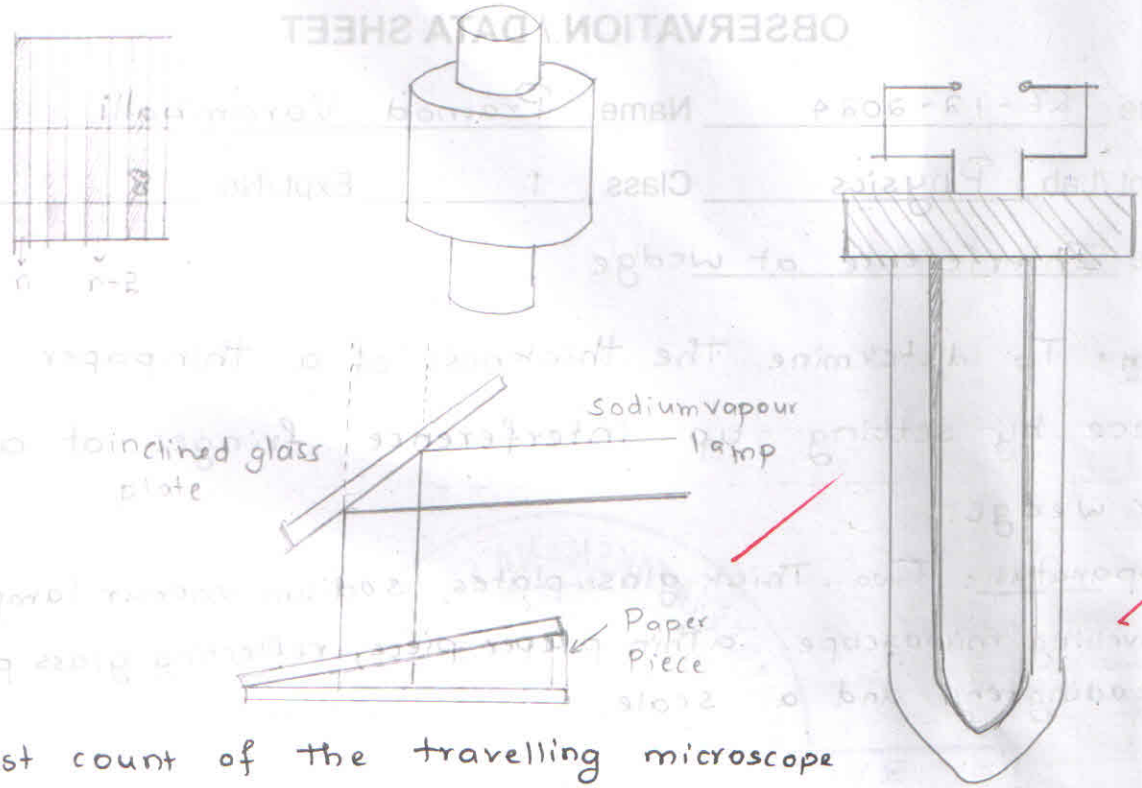
Scheme of Evaluation	Course Outcome	Max Marks	Marks Scored	Initial of the Staff
Data Sheet and Experimental Set up (4+6)	CO1	10	4+6	
Conduction of Experiment	CO2	10	10	
Substitution, Calculation & Accuracy	CO3	10	9	
Total Marks Scored		30	29	
Signature of the Faculty			30	

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Observation



least count of the travelling microscope

$$LC = \frac{\text{Value of main scale division}}{\text{Total number of Vernier Scale division}} = \frac{0.5}{50}$$

Tabular Column

Fringe	TM readings, MSR + (CVD × LC)	Total readings
N	4.4 + (12 × 0.001)	~~4.412~~ 4.412
n+2	4.5 + (5 × 0.001)	~~4.55~~ 4.505
n+4	4.6 + (25 × 0.001	4.625
n+6	4.7 + (04 × 0.001)	4.704
n+8	4.8 + (44 × 0.001)	4.844
n+10	4.95 + (35 × 0.001)	4.985
n+12	5.1 + (18 × 0.001)	5.118
n+14	5.25 + (04 × 0.001)	5.254
n+16	5.35 + (46 × 0.001)	5.396
n+18	5.5 + (31 × 0.001)	5.531
n+20	5.65 + (19 × 0.001)	5.669
n+22	5.8 + (24 × 0.001)	5.824

26/12

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Table 2

Fringe No.	TM Readings	Fringe No	TM Readings	$X = 12\beta = R_2 - R_1$
N	4.412	n+12	5.118	0.706
n+2	4.505	n+14	5.254	0.749
n+4	4.625	n+16	5.396	0.771
n+6	4.704	n+18	5.531	0.827
n+8	4.844	n+20	5.669	0.825
n+10	4.985	n+22	5.824	0.839

= 0.5 / 50 division

The fringe width β, is $X/12 = 0.065513 \times 10^{-2}$ m

To measure The length, L, of air wedge : L = 0.069 m

$t = \frac{\lambda L}{2\beta} = 31.278 \times 10^{-6}$ m

where

t is thickness of the paper 31.278×10^{-6} m

L is length of the air wedge 0.069 m

λ is wavelength of sodium ligth = 589.3 nm

β is the fringe width 0.065513×10^{-2} m

readings
4.412
4.505
4.625
4.704
4.844
4.985
5.118
5.254
5.396
5.531
5.669
5.824

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Calculation

$$t = \frac{\lambda L}{2\beta} = \frac{589.3 \times 10^{-9} \times 0.069}{2 \times 10^{-2} (0.0655)}$$

$$= 312.7823 \times 10^{-7} \text{ m}$$

$$= 31.278 \times 10^{-6} \text{ m}$$

Result :

The thickness of the given paper = 31.278×10^{-6} m

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Fringe pat
system

Inc
pla

Least count of
Va
LC = To

Tabular colu

Fringe numb
N
n + 2
n + 4
n + 6
n + 8
n + 10
n + 12
n + 14
n + 16
n + 18
n + 20
n + 22

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Date _____ Name _____

Dept./Lab _____ Class _____ Expt./No. _____

Title _____

- aim : To determine the moment of inertia of the given irregular body and rigidity modulus of the material of the given wire.

- Apparatus and other material required :- Rectangular circular and irregular shaped plates, steel or brass wire chuck nuts, stopclock, pointer, metrescale, Screwgauge weight box

- Formula

(a) Moment of inertia of irregular body about the axis through the CG and perpendicular to its plane, I'

$$I_\alpha = \left[\frac{1}{T^2}\right]_{mean} \times T_\alpha^2 = \underline{2.8551 \times 10^{-3}}\ kg\ m^3$$

(b) Moment of inertial of irregular body about the axis through the GG and parallel to its plane

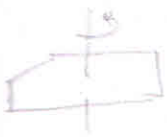
$$I_\beta = \left[\frac{1}{T^2}\right]_{mean} \times T_\beta^2 = \underline{8.6259 \times 10^{-4}}\ kg\ m^3$$

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=) Rigidity modulus $\eta = \left[\frac{8\pi L}{r^4}\right]\left[\frac{I}{T^2}\right]$ where r is the radius and L is the length of the wire.

Body	M (Kg)	Dimension (m)	Axis	Moment of Inertial (Kgm^2)	Time for 10 OSC(s)	mean time t' for 10 cosc	Time Period	$\frac{I}{T^2}$ (Kgm^2)
Circular plate	1.468	radius = 0.0765		$I_1 = \frac{MR^2}{2}$ $= 4.2956 \times 10^{-3}$	1. 43.85 2. 44.22 3. 44.24	1. 2. 3. 44.103	$T_1 = 4.4103$	$\frac{I_1}{T_1^2} =$ 2.20 x
				$I_2 = \frac{mR^2}{4}$ $= 2.1478 \times 10^{-3}$	1. 31.50 2. 31.63 3. 31.20	31.4433	$T_2 =$ 3.1443	$\frac{I_2}{T_2^2} =$ 2.172
Rectangular plate	1.290	Length L = 0.155 Breadth B = 0.045		$I_3 = \frac{M(L^2+B^2)}{12}$ $= 2.7341 \times 10^{-3}$	1. 41.18 2. 40.98 3. 41.09	41.0833	$T_3 =$ 4.1083	$\frac{I_3}{T_3^2}$ = 1.61 x
				$I_4 = \frac{ML^2}{12}$ $= 2.5165 \times 10^{-3}$	1. 33.84 2. 34.09 3. 33.97	33.966	$T_4 =$ 3.3966	$\frac{I_4}{T_4^2}$ = 2.18
				$I_5 = \frac{MB^2}{12}$ $= 0.2176 \times 10^{-3}$	1. 23.22 2. 23.25 3. 23.09	23.1867	$T_5 =$ 2.3187	$\frac{I_5}{T_5^2}$ 0.40 x 10
								= 2.0453

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Axis through CG	Time for 10 sec	Mean time (t) for 10 oscillations (s)	Time period
	1 37.41 2 37.40 3. 37.28	37.3633	$T_\alpha = 3.7363$
	1. 20.60 2. 20.38 3 20.63	20.5367	$T_B = 2.05367$

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Determination body

Formulae

(a) moment o axis throug

$I_\alpha = \left[\frac{I}{T}\right.$

(b) Moment axis throu

$I_\beta = \left[\right.$

Determinat material

Radius of th

Pitch = Di Tot

L.C = Pi T

Zero error=

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Time Period	$\frac{I}{T^2}$ (kgm²)
T_1 = 4.4103	$\frac{I_1}{T_1^2}$ = 2.20 ×
T_2 = 3.1443	$\frac{I_2}{T_2^2}$ = 2.172 =
T_3 = 4.1083	$\frac{I_3}{T_3^2}$ = 1.61 ×
T_4 = 3.3966	$\frac{I_4}{T_4^2}$ = 2.18
T_5 = 3.3187	$\frac{I_5}{T_5^2}$ = 0.40 × 1
= 2.0453	
e period	
3.7363	
= 2.05367	

Determination of Moment of Inertial of Irregular body

Formulae

(a) moment of inertial of irregular body about the axis through the CG and perpendicular to its plane

$$I_\alpha = \left[\frac{I}{T^2}\right]_{mean} \times T_\alpha^2 \quad kgm^2$$

(b) Moment of inertial of irregular body about the axis through the CG and parallel to its plane

$$I_\beta = \left[\frac{I}{T^2}\right]_{mean} \times T_\beta^2 \quad kgm^2$$

Determination of rigidity modulus of the material of the wire

Radius of the wire (r) using screw gauge

Pitch = $\frac{\text{Distance moved on the pitch scale}}{\text{Total number of rotation given}} = \frac{5}{5} = \frac{1}{1}$ mm

L.C = $\frac{\text{Pitch of Screw gauge}}{\text{Total No of head scale divisio}} = \frac{1}{100} = 0.01$ mm

Zero error = -18

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Trial No	PSR (mm)	HSD	TR = PSR + (HSD − ZD) LC mm
1	0	54	0.72
2	0	55	0.73
3	0	56	0.74
4	0	56	0.74

Mean diameter of the wire = $d = 0.7325$ mm $= 0.7325 \times 10^{-3}$ m

mean radius of the wire = $r = 0.3663 \times 10^{-3}$ m

Length of the wire = $L = 0.298$ m

Rigidity modulus

$$\eta = \left[\frac{8\pi L}{r^4}\right]\left[\frac{I}{T^2}\right]$$

Calculations

$I_1 = \frac{MR^2}{2} = \frac{1.468 \times (0.0765)^2}{2} = 4.2956 \times 10^{-3}$ kg m^3

$I_2 = \frac{MR^2}{4} = \frac{1.468 \times (0.0765)^2}{4} = 2.1478 \times 10^{-3}$ kg m^3

$I_3 = \frac{M(L^2+B^2)}{12} = \frac{1.290 \times (0.153^2 + 0.045^2)}{12} = 2.7341 \times 10^{-3}$ kgm^3

$I_4 = \frac{ML^2}{12} = \frac{1.290\,(0.153)^2}{12} = 2.5165 \times 10^{-3}$ kg m^3

$I_5 = \frac{MB^2}{12} = \frac{1.290\,(0.045)^2}{12} = 0.2176 \times 10^{-3}$ kg m^3

$I_d = \left[\frac{I}{T^2}\right]_{mean} \times T_d^2 = 2.0453 \times 10^{-4} \times 3.7363 = 2.8551 \times 10^{-3}$ kgm^3

$I_B = \left[\frac{I}{T^2}\right]_{mean} \times T_B^2 = 2.0453 \times 10^{-4} \times 2.05367 = 8.6859 \times 10^{-4}$ kg m

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Title

. TR = (0) +

. TR = (0) +

. TR = (0) +

. TR = (0)

$= \left[\frac{8\pi L}{r^4}\right]\left[\frac{I}{T}\right.$

$= \frac{8 \times 3.14 \times}{(0.36}$

esult:

The moment

perpendicula

The momen

parallel

Rigidity

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Dept./Lab _____ Class _____ Expt./No. _____

Title _____

1. TR = (0) + (54-(-18)) × 0.01 mm = 0.72 mm = 0.72×10^{-3} m
2. TR = (0) + (55-(-18)) × 0.01 mm = 0.73 mm = 0.73×10^{-3} m
3. TR = (0) + (55-(-18) × 0.01 mm = 0.74 mm = 0.74×10^{-3} m
4. TR = (0) + (56-(-18) × 0.01 mm = 0.74 mm = 0.74×10^{-3} m

$d_{mean} = 0.7325 \times 10^{-3}$ m

$$= \left[\frac{8\pi L}{r^4}\right]\left[\frac{I}{T^2}\right]_{mean}$$

$$= \frac{8 \times 3.14 \times 0.298}{(0.3663 \times 10^{-3})^4} \times 2.0453 \times 10^{-4} = 2.8553 \times 10^{11} \text{ N/m}^2$$

$$= 8.5087 \times 10^{10} \text{ N/m}^2$$

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Result:

The moment of Inertial of Irregular body about axis perpendicular to the plane $I_\alpha = 2.855 \times 10^{-3}$ kgm²

The moment of Inertial of Irregular body about axis parallel to the plane $I_\beta = 8.6259 \times 10^{-4}$ kg m²

Rigidity modulus of material of wire = $\eta = 2.8553 \times 10^{11}$ N/m²

$= 8.5087 \times 10^{10}$

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Aim:- To determine the wavelength of a given laser b

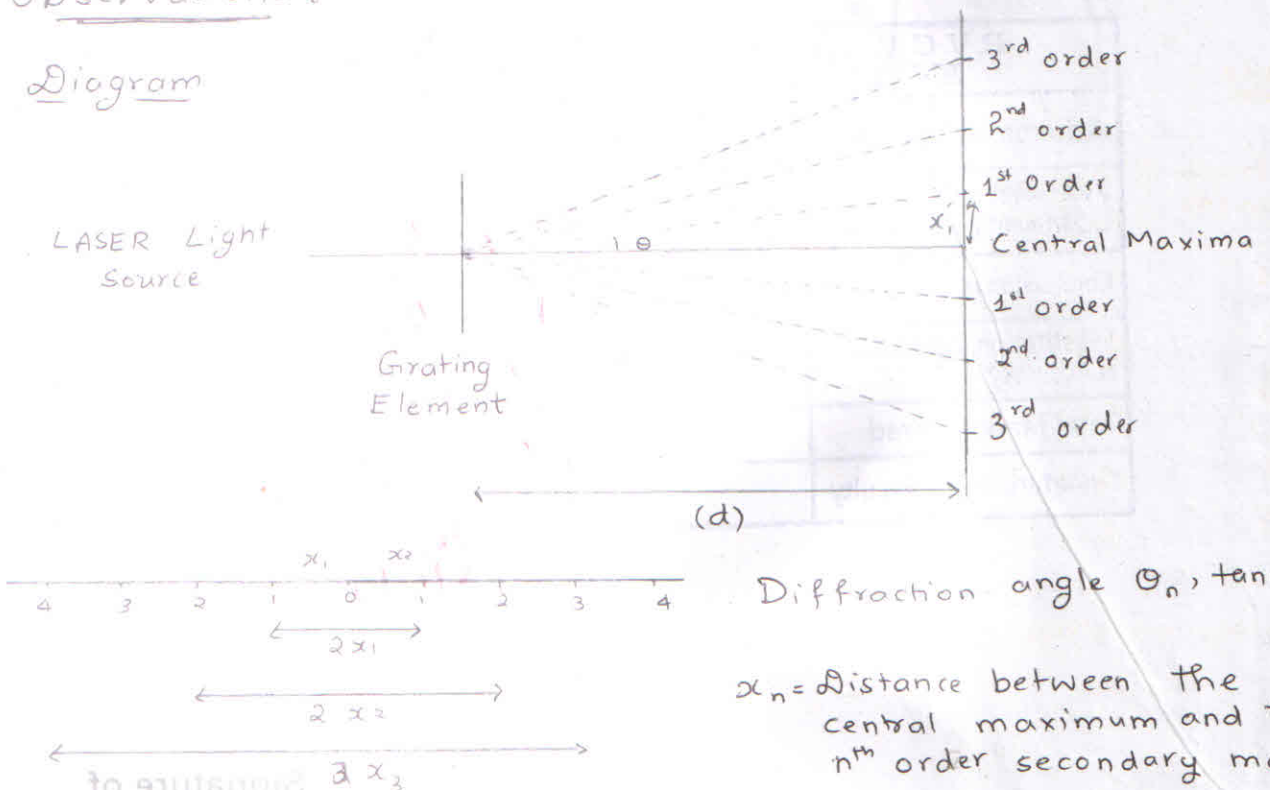
Apparatus:- Laser source, Grating, Optical bench with accesso
and metre scale etc.

Formula:- Wavelength of lazer light, $\lambda = \frac{C \sin\theta_n}{n}$ ..... m

where 'C' is the grating constant and it is the distance between corresponding point of two successive lines on the grating. n is the order of the maximum, θ_n is the angle of difraction of nth maximum.

Observations:

Diagram



Diffraction angle θ_n, tanθ

x_n = Distance between the central maximum and th n^{th} order secondary ma

d = Distance between grat and the screen

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Title

ormula

Wavelength

where 'C' i
the maximu

Grating Cons

distance betu

ble:

ffraction den (n)	Distanc 2 Xn (c
1	3.3
2	5.2
3	10.1
4	13.3
5	16.8

verage wavelen

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Title _____

Formula

Wavelength of laser source $\lambda = \frac{C \sin\theta_n}{n}$ = _____ m

where 'C' is the grating constant, n is the order of the maximum, θ is the angle of diffraction

Grating Constant: $C = \frac{1 \text{ inch}}{\text{No. of lines (N) per inch}} = \frac{2.54 \times 10^{-2}\ m}{500} = 5.08 \times 10^{-5}\ m$

distance between the grating and the screen, d = 121.8 cm

Table:

Fraction den (n)	Distance $2X_n$ (cm)	Distance X_n (cm)	Diffraction angle (θ_n in $\theta_n = \tan^{-1}(\frac{x_n}{d})$)	Wavelength λ (nm) $\lambda = \frac{C \sin\theta_n}{n}$	Fractional error $(\frac{\delta\lambda}{\lambda})$
1	3.3	1.65	0.~~9~~761	~~0.0~~688.8903	0.06~~0~~6
2	5.2	~~2.6~~ 3.1	1.4~~2~~79	~~0.0~~646.2372	0.0322 ~~0.0526~~
3	10.1	5.05	2.3742	~~0.0~~701.4760	0.0198 ~~0.0229~~
4	13.3	6.65	3.1251	~~0.0~~692.3561	0.0150 ~~0.0096~~
5	16.8	8.4	3.9516	700.6387	0.0119 ~~0.0216~~
Average wavelength of laser and fractional error				λ = 585.7594 nm	$(\frac{\delta\lambda}{\lambda})$ = 0.0279

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Error Formula

The formula used wavelength of laser light $= \lambda = C s$

Taking the logarith, $\log \lambda = \log\left(\frac{c}{n}\right) + \log(\sin\theta)$

con sympliffing

$$\frac{\delta\lambda}{\lambda} = \frac{\delta x_n}{x_n} + \frac{x_n \delta x_n + d\,\delta d}{(x_n^2 + d^2)} = \frac{\delta x_n}{x_n}$$

Calculations

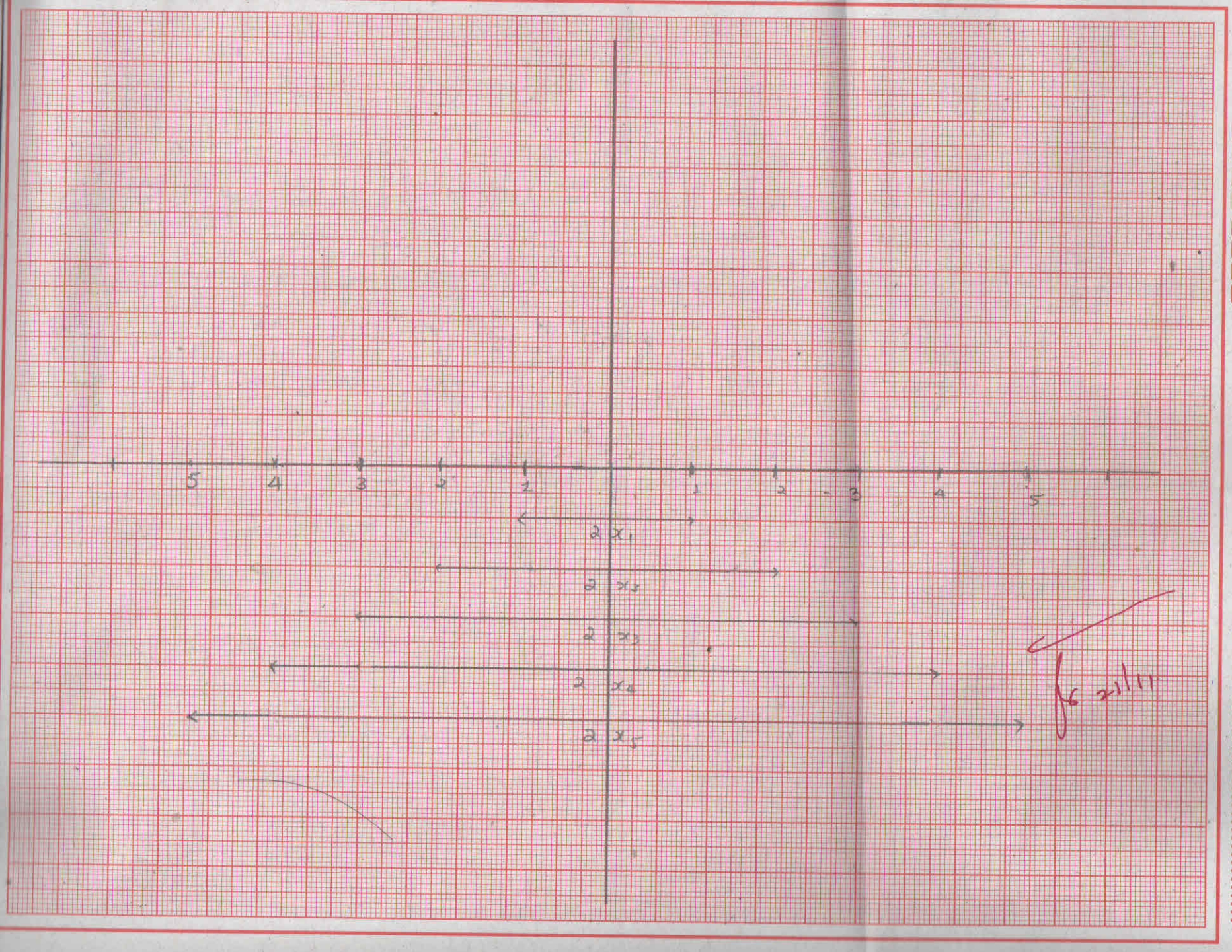
1. $\theta_1 = \tan^{-1}\left(\frac{1.65}{121.8}\right) = 0.9761$ $\quad \lambda_1 = \frac{C \sin\theta_1}{1} = 688.0903$ nm
2. $\theta_2 = \tan^{-1}\left(\frac{3.1}{121.8}\right) = 1.4519$ $\quad \lambda_2 = \frac{C \sin\theta_2}{2} = 646.2372$ nm
3. $\theta_3 = \tan^{-1}\left(\frac{5.05}{121.8}\right) = 2.3742$ $\quad \lambda_3 = \frac{C \sin\theta_3}{3} = 701.4760$ nm
4. $\theta_4 = \tan^{-1}\left(\frac{6.65}{121.8}\right) = 3.1251$ $\quad \lambda_4 = \frac{C \sin\theta_4}{4} = 692.3561$ nm
5. $\theta_5 = \tan^{-1}\left(\frac{8.4}{121.8}\right) = 3.9516$ $\quad \lambda_5 = \frac{C \sin\theta_5}{5} = 700.6387$ nm

Result :- 1. The wavelength of lazer light is found to be 685.759

2. The relative error (or fractional error) in λ is 0.0279

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Batch No.....



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OBSERVATION / DATA SHEET

Date 29.11.2024 Name Atharva Somani

Dept./Lab IEM/Physics Lab-1 Class _____ Expt./No. _____

Title Volume Resonator

Aim- To determine the frequency of a tuning fork using volume resonator.

Apparatus- Aspirator bottle, tuning forks, rubber pad, beaker, measuring jar.

Formula- $f_{unknown} = \sqrt{\frac{[f^2(V+\Delta V)_{avg}]}{V_f + \Delta V}}$ H_2

where $f_{unknown}$ is the unknown frequency of the tuning fork, f = known frequency of the tuning fork, V = volume of air resonating with the fork of known frequency, V_f = volume of ~~known~~ air resonating with the fork of unknown frequency and ΔV is the volume correction obtained from graph.

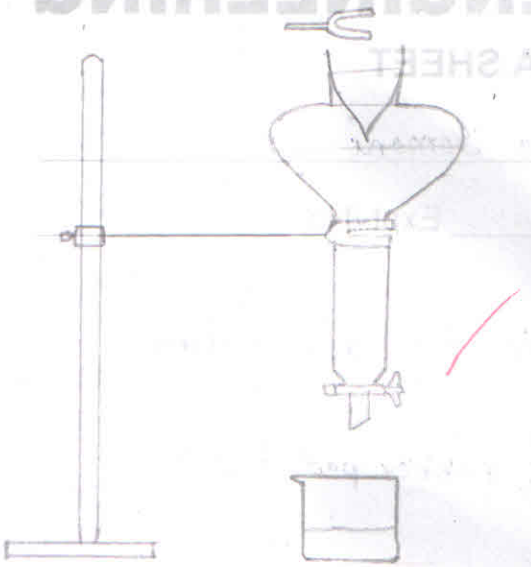
Particulars	Max Marks	Marks Obtained	Faculty Signature
Data Sheet + Experimental setup + Viva Voce	10	4+6	
Conduction of Experiment	10	10	
Substitution + Calculation + Accuracy	10	9	
Total Marks	30	29/30	

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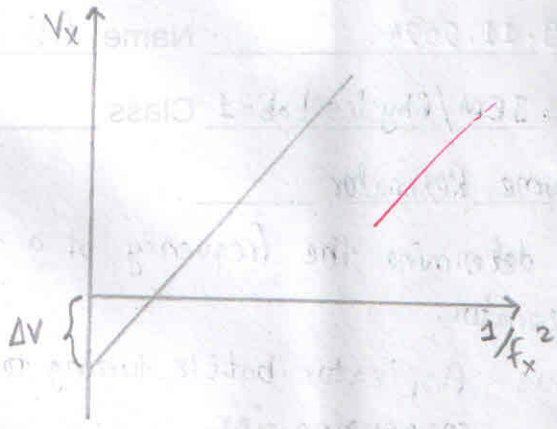
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Observations :-

Experimental Setup :-



Model Graph :-



Sl. No.	Tuning Fork Frequency f in Hz	Volume of resonating air (water collected) in cm^3			$\frac{1}{f^2}$ (10^{-6})	$f^2.(V+\Delta V)$ in $cm^3 s^{-2}$	$[f^2(V+\Delta$
		V_1 in cm^3	V_2 in cm^3	Mean Volume V in cm^3			
1.	480	154	150	152	4.3403	32256000	
2.	426	192	194	193	5.5104	32847156	314823
3.	384	212	210	211	6.7817	29343744	
4.	funknown	136	132	V_f 134	-	$(V_f+\Delta V)$ = 122	-

Determining unknown frequency of the tuning fork

$$[f^2(V+\Delta V)]_{avg} = f^2_{unknown}\,(V_f+\Delta V)_{unknown}$$

$$f_{unknown} = \sqrt{\frac{[f^2(V+\Delta V)_{avg}]}{V_f+\Delta V}}\ Hz = 507.9878\ Hz$$

where, $f_{unknown}$ is the unknown freq. of the tuning fork, f = Known freq. of tuning fork, V = vol. of air resonating with the fork of known freq., V_f of air resonating with the fork of unknown freq. and ΔV is the volu correction obtained from graph.

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sult :-

ume correction =

unknown freque

lculations :-

$(V+\Delta V)$

$80^2 (152-12) =$

$26^2 (193-12) =$

$84^2 (211-12) =$

nknown = $\sqrt{[f^2(V+$

$(V_f$

$= \sqrt{\frac{31482}{12}}$

$= 507.98$

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e _____ Name Atharva Somani

ot./Lab _____ Class _____ Expt./No. _____

e _____

sult:-

ume correction = $\Delta V = -12$

unknown frequency of given tuning fork = ~~Hz~~

$= 507.9878$ Hz

lculations :-

$(V + \Delta V)$

$80^2 (152 - 12) = 32256000 \text{ cm}^3 s^{-2}$

$26^2 (193 - 12) = 32847156 \text{ cm}^3 s^{-2}$

$84^2 (211 - 12) = 29343744 \text{ cm}^3 s^{-2}$

$$\text{nknown} = \sqrt{\frac{[f^2(V+\Delta V)]_{avg}}{(V_f + \Delta V)}} \text{ Hz}$$

$$= \sqrt{\frac{31482300}{122}}$$

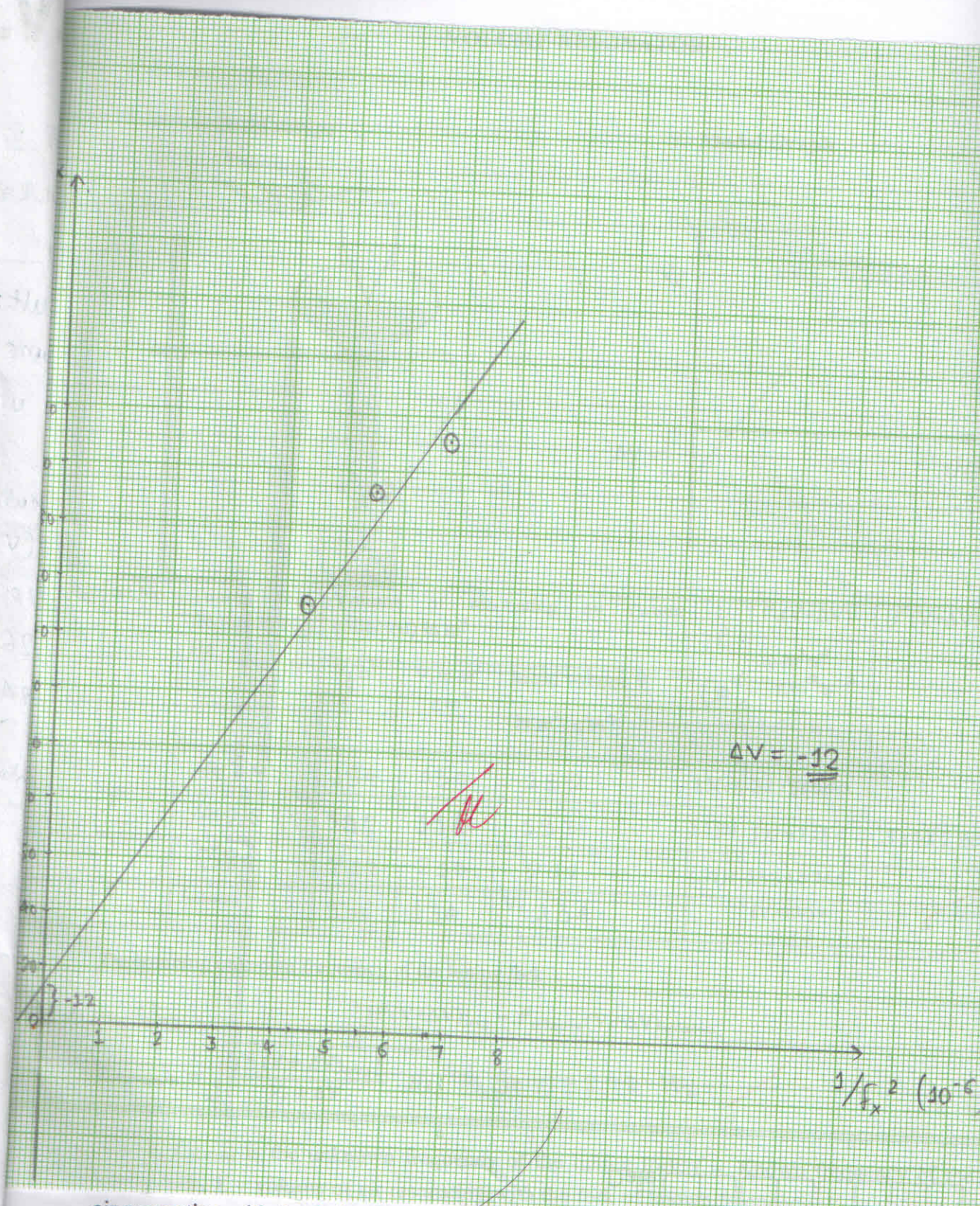
$= 507.9878$ Hz

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$\Delta V)$ in s^{-2}	$[f^2(V+\Delta$
00	
56	314823
44	
	-

Known freq. of
own freq., V_f
ΔV is the volu

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air resonating with the fork of unknown frequency and ΔV is the volume correction obtained from graph.

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OBSERVATION / DATA SHEET

Date 28-11-2024 Name Pramod Varamballi

Dept./Lab Physics - 2 Class _____ Expt./No. 5

Title Band Gap of thermistor.

Aim. To determine the energy gap (E_g) of a thermometer

Apparatus: Glass beaker, Thermistor, multimeter Thermometer.

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R V C E, DEPARTMENT OF ...
EVALUATION OF ...

Scheme of Evaluation	Course Outcome	Max Marks	Marks Scored	Initials of Staff
Data Sheet and Experimental Set up (4+6)	CO1	10	4+6	
Conduction of Experiment	CO2	10	10	
Substitution, Calculation & Accuracy	CO3	10	10	
Total Marks Scored		30	30	
Signature of the Faculty			30	

Formula = $\frac{4.606 \times K \times m}{1.6 \times 10^{-19}}$ eV

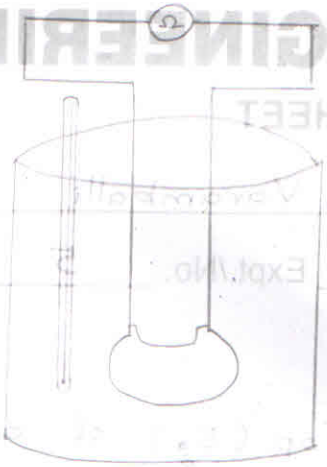
E_g = Energy gap of given thermistor in eV.

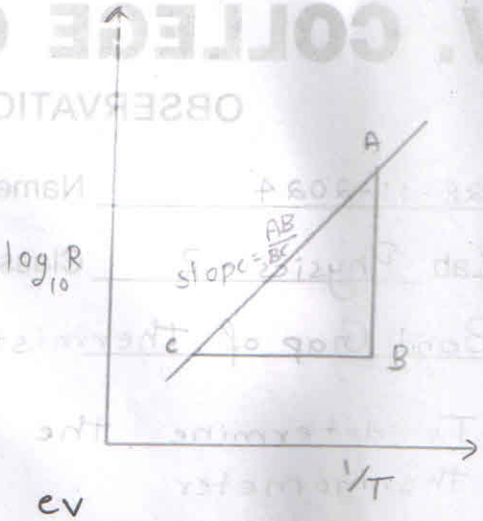
$K = 1.381 \times 10^{-23}$ J/K

m = slope of graph of log R vs 1/T

Signature of Teacher Incharge

Observation





Formula : $E_g = \frac{4.606 \times K \times m}{1.6 \times 10^{-19}}$ ev

E_g = energy gap of a given thermistor in ev

k = Boltzmann constant = 1.381×10^{-23} J/k

m = slope of graph

Table

Sl. No	Temp t °C	Temp T(K)	R (Ω)	log R	1/T ($\times 10^{-3}$ K^{-1})
Room temp	27.8	300.15	53.1 k	1.7251	3.3316
1	81	354.15	8.4	0.9243	2.8236
2	80	353.15	8.6	0.9345	2.8316
3	79	352.15	8.8	0.9445	2.8396
4	78	351.15	9.1	0.9590	2.8478
5	77	350.15	9.3	0.9684	2.8559
6	76	349.15	9.6	0.9823	2.8641
7	75	348.15	9.8	0.9912	2.8723
8	74	347.15	10.1	1.0043	2.8806
9	73	346.15	10.5	1.0211	2.8889
10	72	345.15	10.8	1.0334	2.8972

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Date

Dept./Lab

Title

Calculation

slope = AB/BC

E_g = 4.606 × 1

1.

= 5.66

esult: The e

thermis

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OBSERVATION / DATA SHEET

Date _____ Name _____

Dept./Lab _____ Class _____ Expt./No. _____

Title _____

Calculation

slope = $\frac{AB}{BC} = \frac{5.7}{4} = 1.425$

$E_g = \frac{4.606 \times 1.381 \times 10^{-23} \times 1.425 \times 10^{3}}{1.6 \times 10^{-19}}$ eV

$= 5.665 \times 10^{-4}$ eV

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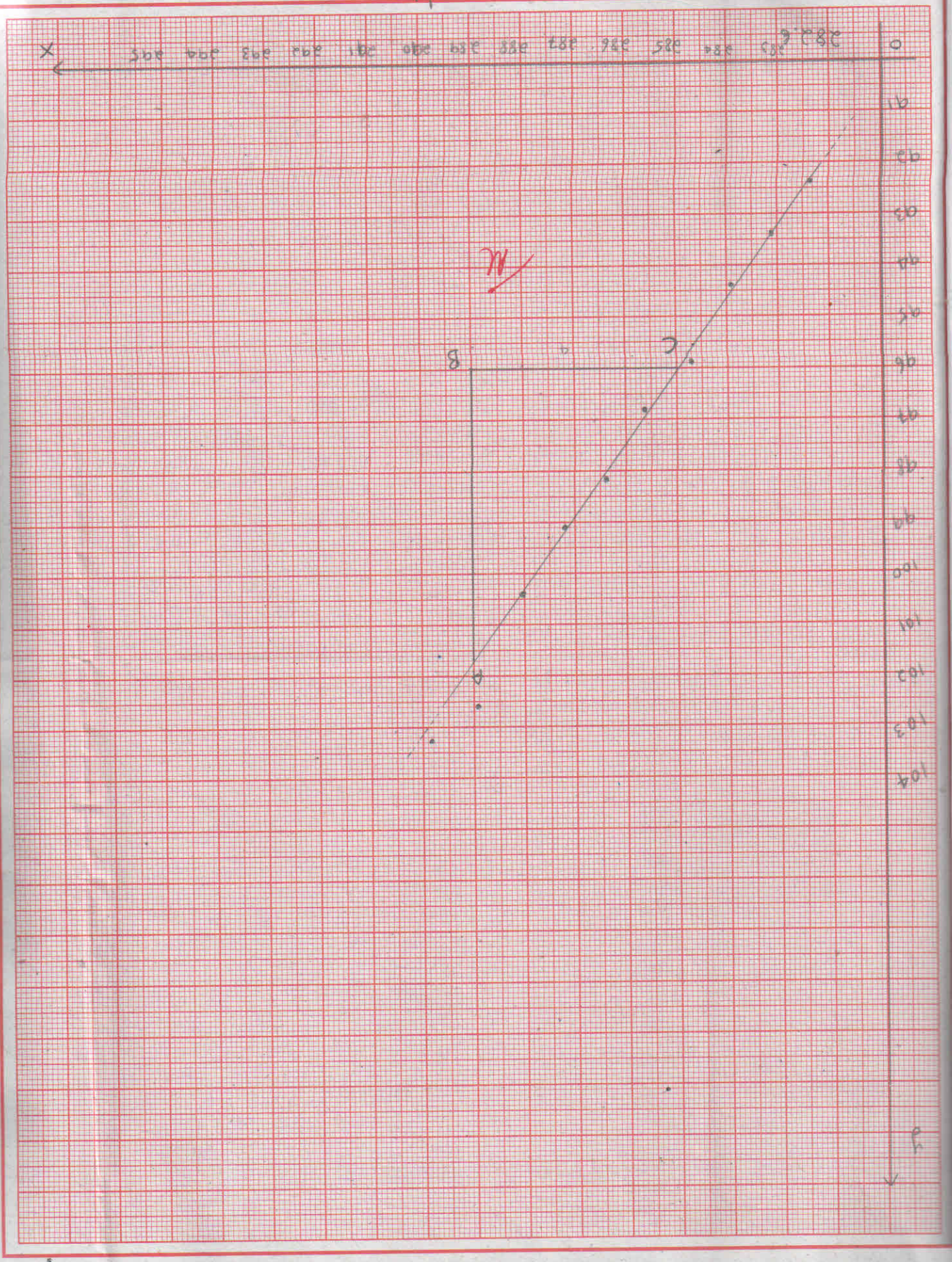
Result: The energy of gap of the given thermister is 5.665×10^{-4} eV

Signature of Teacher Incharge

1/T ($\times 10^{-3}$ K^{-1})
3.3316
2.8236
2.8316
2.8396
2.8478
2.8559
2.8641
2.8723
2.8806
2.8889
2.8972

DATE : ☐☐☐☐☐☐☐☐

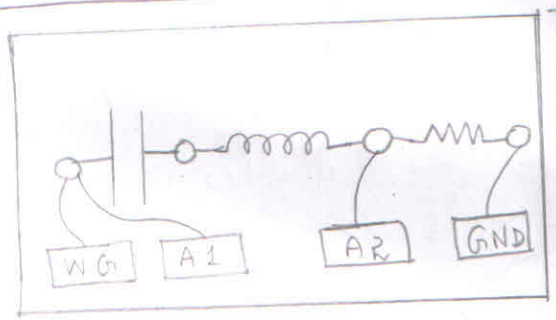
Batch No.....

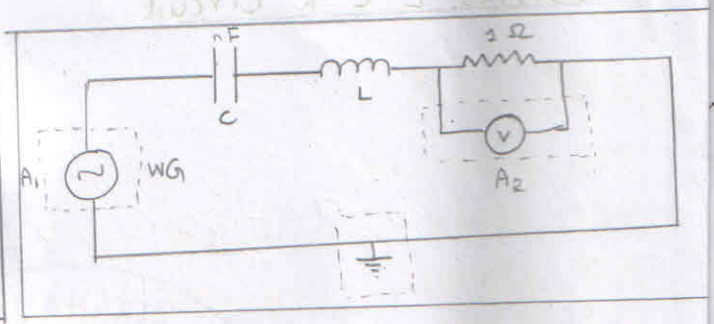


Aim: To study the frequency response of LCR circuit and determination of a) self inductance of the given coil, b) Quality factor, c) Bandwidth

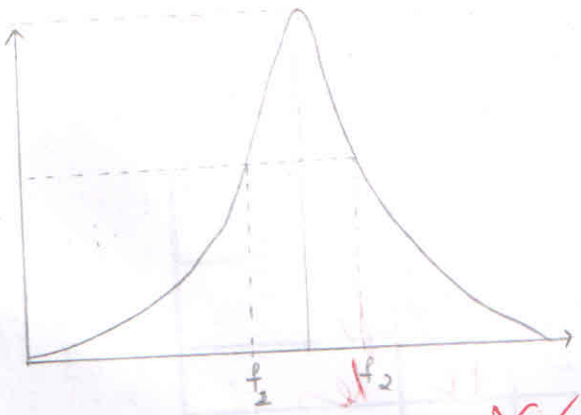
Apparatus: expEYES-17 kit, Resistance box, Capacitor, Inductor.

Observations





Model Graph



Formulae

Capacitance C = 474 nF

$L = \frac{1}{4\pi^2 f_0^2 C}$ = 1.1035 H

I_{max} = ~~17.1878~~ 21.9449 mA

$I_{RMS} = \frac{I_{max}}{\sqrt{2}}$ = ~~12.1536~~ 15.51738 mA

Resonant Frequency = f_0 = ~~220.064~~ 220.064 Hz

Lower cut off frequency f_1 = 204.165 Hz

Upper Cut off frequency f_2 = 235.575 Hz

Band width $\Delta f = f_2 - f_1$ = H

Quality factor = Q = $\frac{f_0}{\Delta f}$ =

Calculations:

$L = \frac{1}{4\pi^2 (220.064)^2 \times 474 \times 10^{-9}}$ = 1.1035 H

Band width = $f_2 - f_1$ = 235.575 - 204.165 = 31.41 Hz

Quality factor = $\frac{f_0}{\Delta f} = \frac{220.064}{31.41}$ = 7.00617

R.V.

Date

Dept./Lab

Title

Sample

set ti
set t
set x
set y
plot

esult :

. Resonan
. Self ind
. Quality
. Bandw

SERIES L-C-R CIRCUIT

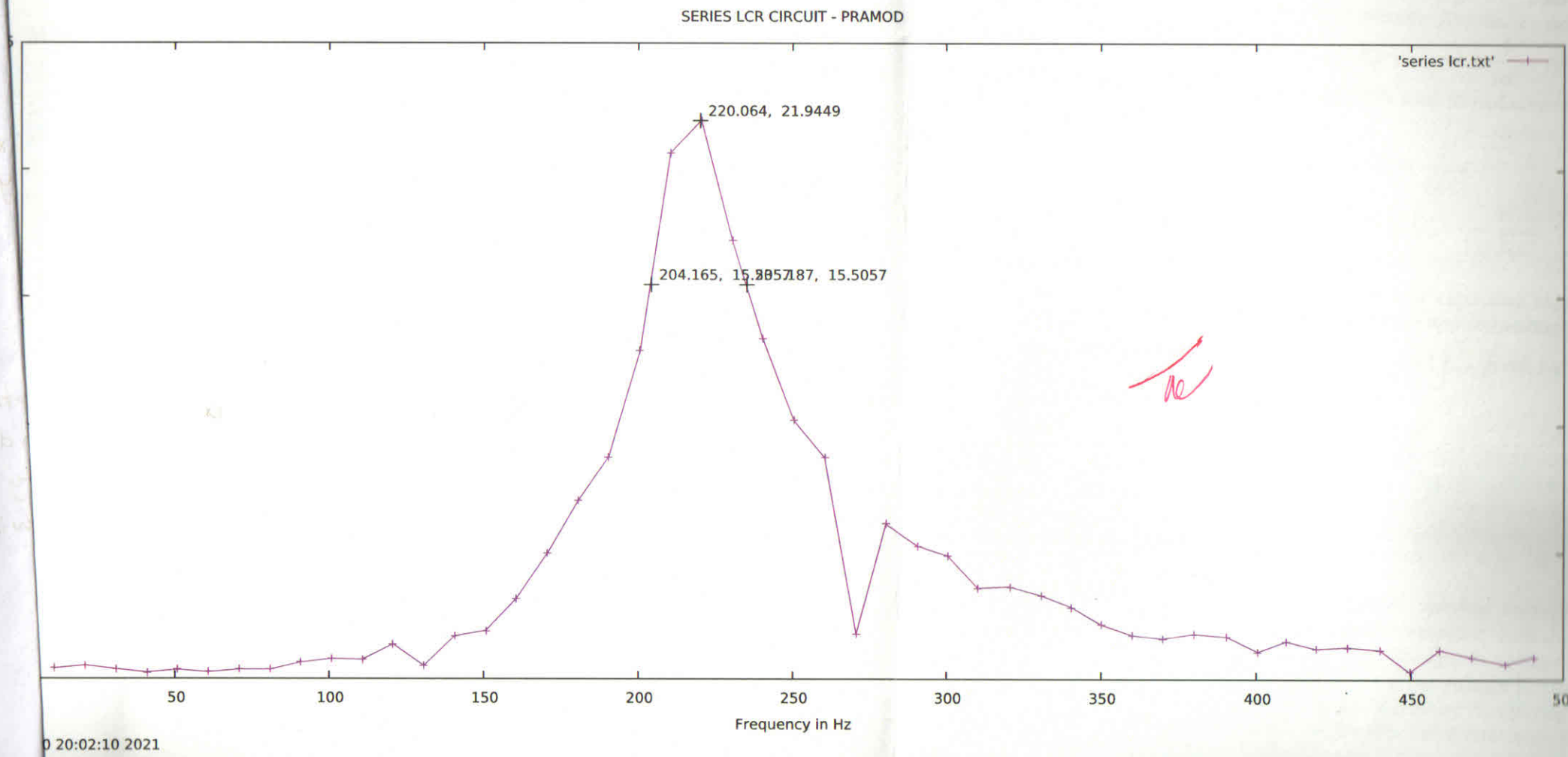
OBSERVATIONS:

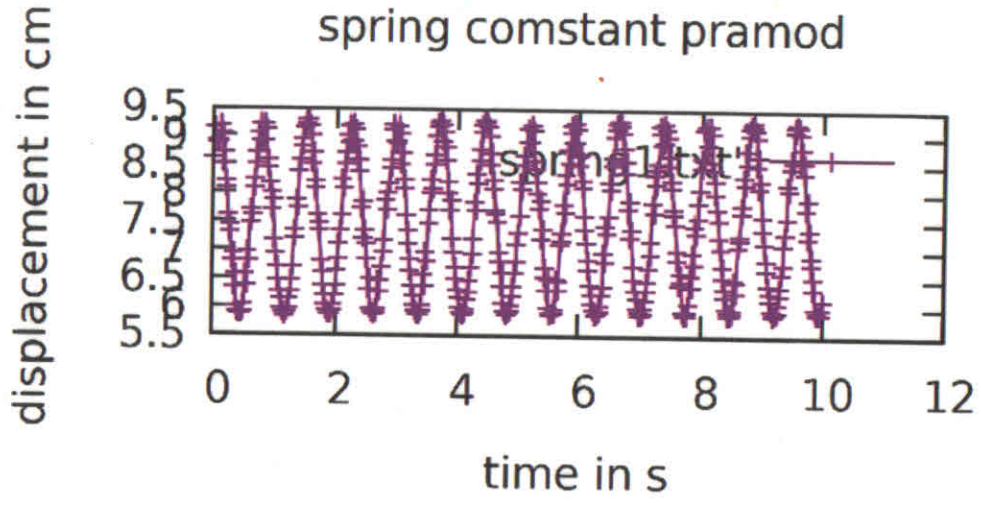
ExpEyes circuit:

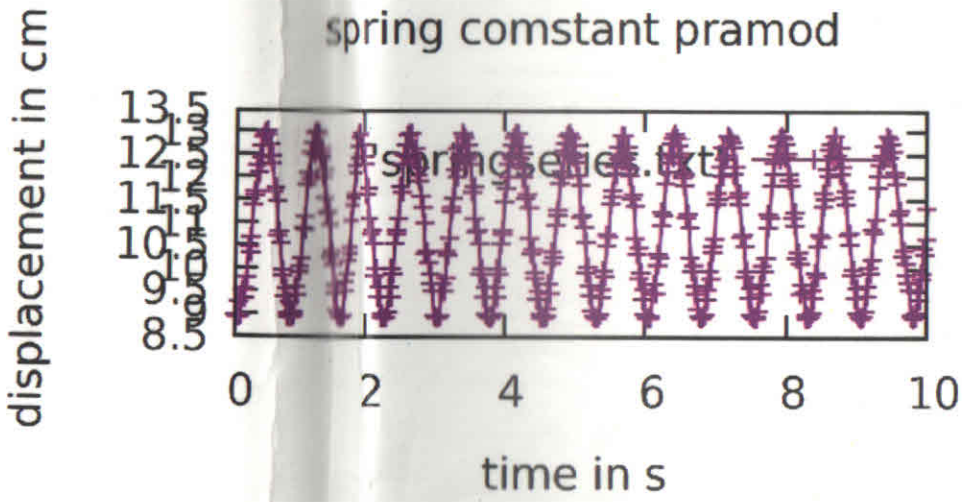
Conventional Circuit:

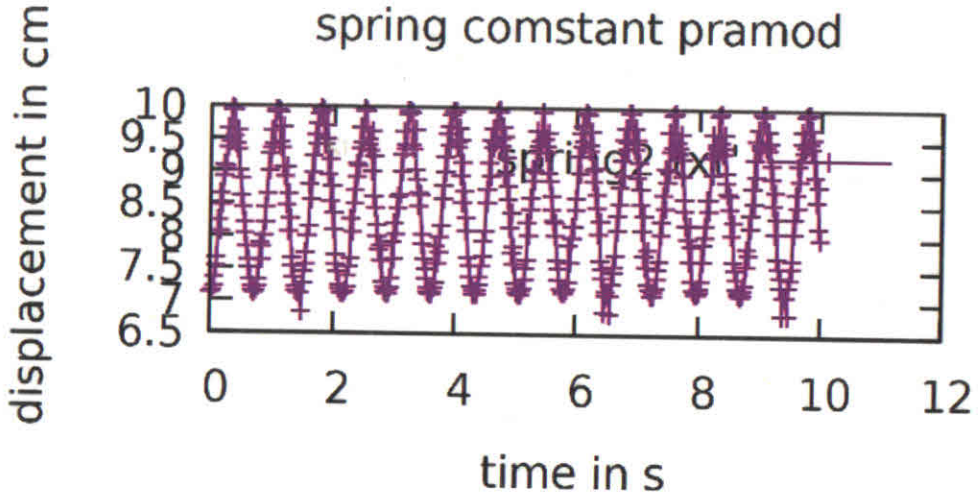
SERIES L-C-R CIRCU

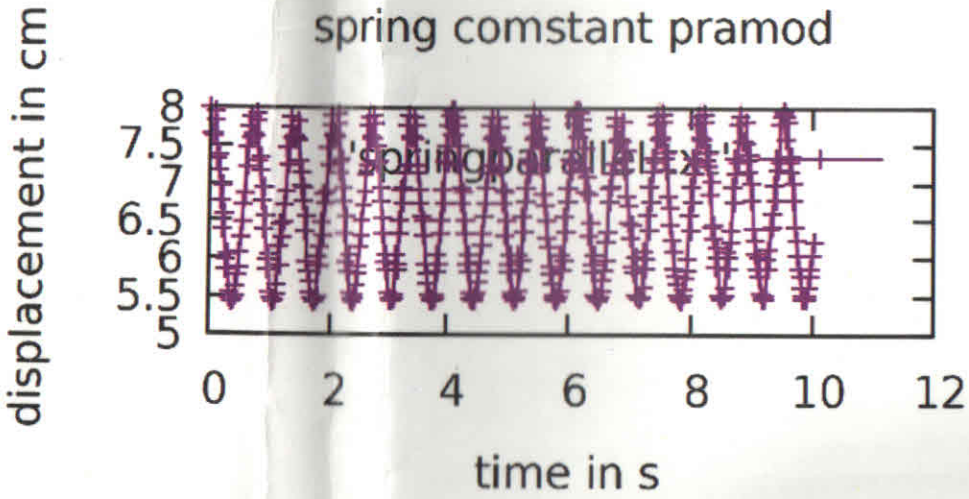
Experiment No: 4











Thu Nov 14 04:45:06 2024